5SWB0 Advanced Power Electronics · EMC lecture demonstrator

# EMC Demonstration Flyback Converter

## Design report: 24 V → 12.2 V / 1 A isolated flyback with switchable EMC knobs

| Item | Value |
|---|---|
| Document | Design record, board revision E (100 × 76 mm, aluminium case as the Q1 heatsink, compact HF power cell) |
| Author | Chengmin Li, Electromechanics and Power Electronics (EPE), TU/e |
| Date | 3 October 2026 |
| Status | Design complete, verified by simulation. **Not yet prototyped.** |
| Design files | KiCad project, Gerbers and BOMs: EMC_demo_flyback / 2_slides\EMC Improve<br>Simulation: 2_slides\EMC Improve\simulation<br>Topology drawing: EMC_flyback_topology.drawio<br>Case: EMC_demo_flyback\case (drawing, STEP) |

# Contents


**1 Summary ..... 3**

**2 Purpose, concept and specifications ..... 4**

2.1 Purpose ..... 4
2.2 EMC knobs ..... 4
2.3 Specifications ..... 5

**3 Circuit topology ..... 6**

**4 Parameter design ..... 7**

4.1 Switching frequency and controller ..... 7
4.2 Operating mode and transformer ..... 7
4.3 Semiconductor stresses and selection ..... 8
4.4 RCD clamp ..... 9
4.5 RC snubber knob and secondary snubber ..... 9
4.6 Gate drive and the Rg knob ..... 9
4.7 Current sensing and protection ..... 10
4.8 Output filter ..... 10
4.9 Input capacitors and EMI filter ..... 10
4.10 Feedback and compensation ..... 11
4.11 Auxiliary supply and indicators ..... 12
4.12 Losses and thermal design ..... 12

**5 Simulation ..... 13**

5.1 Models and set-up ..... 13
5.2 Steady-state waveforms ..... 14
5.3 Rg knob and RC snubber: edges and spectrum ..... 15
5.4 Losses and efficiency ..... 16
5.5 Closed-loop regulation and load steps ..... 16
5.6 Cold start ..... 17
5.7 Calculation versus simulation ..... 18
5.8 Thermal estimate of the hot components ..... 18

**6 PCB implementation and mechanics ..... 21**

6.1 Layout ..... 21
6.2 Aluminium case (Q1 heatsink) and assembly ..... 24
6.3 The case as an EMC knob ..... 28

**7 Key components ..... 31**

**8 Bring-up and test plan ..... 31**

8.1 Bring-up ..... 31
8.2 EMC measurements: set-up ..... 32
8.3 EMC measurements: procedure ..... 33

**9 Risks and open points ..... 35**

**Appendix A Design files ..... 36**

**Appendix B Bill of materials ..... 36**

# 1 Summary

This report documents the design of a small isolated flyback converter built as a live demonstrator for the EMC lectures of 5SWB0. The converter turns a 24 V bench supply into 12.2 V at 1 A. Its EMC-relevant details can be changed with jumpers during the lecture: the common-mode and differential-mode filter stages, the Y-capacitors, the MOSFET gate resistance (switching speed), a drain RC snubber, the potential of the aluminium case that is also the MOSFET heatsink, the primary-to-secondary Y-capacitor and the output earthing. Conducted emissions are measured through two 5 µH LISNs.

The power stage is a discontinuous-conduction-mode (DCM) flyback with peak-current-mode control: a UC3845B controller at 153.6 kHz drives an IRF520N MOSFET through a Würth 750311591 transformer (8 µH, 1.5:1). An RCD clamp limits the drain voltage to about 66 V. A TL431 and an LTV-817S optocoupler close the loop across the isolation barrier. The board is a two-layer, 100 × 76 mm PCB. It sits on 8 mm stand-offs in a half-open, bent aluminium U-channel case. Q1 is mounted under the board with its tab on the case floor, so the case is the heatsink, as in the heatsink-and-housing example of the lecture. Right-angle 4 mm sockets pass through slots in the two end walls.

The design was checked against hand calculations and an LTspice schematic, which was verified in ngspice because LTspice was not available. Table 1 summarises the main results.

**Table 1.** Key results at 24 V input, 1 A load, 25 °C (simulation unless stated).

| Quantity | Result | Requirement / comment |
|---|---|---|
| Output voltage | 12.22 V (1 A), 12.24 V (0.5 A) | Set point 12.23 V (TL431, R15/R16) |
| Output ripple | 75 mV peak-to-peak | ≤ 100 mV |
| Load step 1 A ↔ 0.5 A | ±22 mV, settled in ~40 µs, no ringing | Stable |
| Loop crossover / phase margin | 5.2 kHz / 73° (CTR 1)<br>2.7 kHz / 80° (CTR 0.5) · 9.4 kHz / 63° (CTR 2) | PM > 45° over the opto CTR spread (small-signal model) |
| Efficiency (power stage) | 83.5 % (Rg 4.7 Ω) · 82.2 % (22 Ω) · 77.3 % (100 Ω)<br>board level ≈ 78–79 % at Rg 4.7 Ω | ≥ 80 % at the default setting |
| MOSFET drain peak | 66 V (24 V in), ~73 V at 30 V in (calc.) | IRF520N 100 V: 27 % margin |
| Primary peak current | 4.73 A steady state, ≈ 8.5 A first cycles of a cold start | T1 $I_{sat}$ = 20 A; current limit 6.7 A nominal |
| Drain dv/dt (Rg knob) | 5.0 / 2.5 / 0.65 V/ns for 4.7 / 22 / 100 Ω | Main EMC knob |
| Hottest parts (board thermal model, 25 °C) | R9/R10/R20 70 °C · D1 68 °C · R4 53 °C · D3/D7 51 °C<br>Q1 junction 41 / 45 / 58 °C for 4.7 / 22 / 100 Ω, case 34–40 °C | All within ratings (Section 5.8) |
| Q1 tab-to-case capacitance | ≈ 30 pF (Sil-Pad 400) → CM current peaks of 150 / 75 / 20 mA for Rg 4.7 / 22 / 100 Ω | The case is a CM knob (CASE), Section 6.3 |

**Main findings for the hardware.** (1) The loop is stable over the full optocoupler CTR spread and the 18–30 V input range. (2) The RCD clamp, not the MOSFET, is the largest single loss (0.91 W at 24 V, 1 A); together with Q1 it sets the efficiency. (3) In simulation the leakage-inductance ringing after turn-off dominates the switch-node spectrum above 15 MHz. Slowing the edge with Rg mainly lowers the 5–15 MHz band, whereas the RC snubber (SNUB) removes the 20–30 MHz ringing peak (≈ 20 dB). This should be confirmed with LISN measurements. (4) There is no soft-start: a cold start runs at the current limit for ≈ 2.7 ms. The 8.5 A peak is well below the transformer saturation current. (5) Thermally, rev C spread the clamp loss over three 4.7 kΩ resistors on local copper (≈ 92 → 68 °C) and added D7 in parallel with D3. Rev D moves Q1 onto the aluminium case: Q1 runs at ≈ 41 °C (case ≈ 34 °C) and still below 70 °C at Rg = 100 Ω. The hottest parts are the clamp resistors (≈ 70 °C) and D1 (≈ 68 °C). (6) The simulated 83.5 % is the power-stage efficiency; with the input diode, the L78L12 supply, the LEDs and the preload the board-level efficiency at 1 A is ≈ 78–79 %. (7) The case is part of the common-mode path. Through the ≈ 30 pF tab capacitance alone, with CASE = CHASSIS and without the EMI filter, the estimated LISN voltage is ≈ 75–100 dBµV

from 150 kHz to 30 MHz, i.e. 20–40 dB above the class B limit. CASE = PGND or 1 nF returns this current locally, which makes the case knob one of the strongest demonstrations. (8) Rev E shrinks the high-frequency loops of the power cell (Section 6.1): the clamp and snubber now act within ≈ 13–14 nH of the drain instead of 26–32 nH, so the turn-off spike that the clamp does not catch and the ringing that the snubber damps are both smaller and better defined.

**Table 2.** Revision history.

| Rev | Date | Change |
|---|---|---|
| A | Oct 2026 | First layout, 130 × 96 mm |
| B | 2 Oct 2026 | Compact 100 × 76 mm layout, same power-stage core and hot loops; Fischer SK 104 25.4 mm heatsink |
| C | 3 Oct 2026 | Thermal improvements: clamp resistor split over R9/R10/R20 (3 × 4.7 kΩ 1 W, was 2 × 3.0 kΩ) on local CLAMP/VBUS copper with thermal vias; second rectifier diode D7 in parallel with D3 with shared VOUT copper. Clamp resistors ≈ 92 → 68 °C, rectifier 57 → 49 °C |
| D | 3 Oct 2026 | Aluminium U-channel case as the Q1 heatsink: Q1 under the PCB (leads bent 90°, tab on the case floor with a Sil-Pad 400, M3 screw through a 7 mm PCB access hole); SK 104 heatsink removed; metal stand-off MH5 connects the case to the CASE knob, MH1–MH4 nylon (no copper); right-angle 4 mm sockets (Cliff FCR7350) through open slots in the case walls; Q1 1 mm lower, J1–J5 re-placed. No circuit change |
| E | 3 Oct 2026 | High-frequency layout of the power cell: D2/C15 (RCD clamp) moved next to T1 pins 1-4, C5/C6 and Q1; R4 directly under the Q1 source; RC snubber SNUB/C16/R11 moved from 25 mm away to between Q1 drain and R4 → PGND; D3/D7 and C20/C21 moved next to T1 pins 7-12 with a straight SGND return strip. Estimated loop inductance: clamp commutation 26 → 14 nH, snubber 32 → 13 nH, primary power loop 25 → 18 nH, secondary 27/34 → 15/25 nH (Section 6.1). Q1 tab hole and case drawing moved with Q1. Voltage-based clearance/creepage rules in the KiCad project (2.5 mm primary–secondary, 1.5 mm chassis–secondary, 0.6 mm around the 100 V nets) with local re-routes to meet them. Jumpers and test points renamed by function (JP1–JP9 → CM1, CM2, YCAP, DM, RG, CASE, SNUB, SGCH, YPS; TP1–TP10 → DRAIN, GATE, CS, PGND1, PGND2, VOUT, SGND, HS, VBUS, CHAS). No circuit change |

# 2 Purpose, concept and specifications

## 2.1 Purpose

The board must show, on a real converter, how layout and component choices change conducted (and near-field) emissions. It follows Tim Williams' heatsink experiment (Elmac, *Characterisation of emissions due to power electronics heatsinks*), but uses a working flyback instead of an oscillator with a resistor load. It also covers the flyback part of Exercise 11.2 (primary-secondary capacitance and the Y-capacitor). Every knob must be switchable in seconds during a lecture, and the measurement chain (two 5 µH LISNs, an optional CM/DM separator, a spectrum analyser or scope FFT) must be standard lab equipment.

## 2.2 EMC knobs

From rev E the jumpers and test points carry functional reference designators instead of JP1–JP9 and TP1–TP10, so the silkscreen says what each one does: CM1/CM2 (CM choke bypass), YCAP, DM, RG, CASE, SNUB, SGCH (SGND to chassis) and YPS (primary–secondary Y-capacitor); test points DRAIN, GATE, CS, VBUS, HS (case), PGND1/2, VOUT, SGND and CHAS (chassis).

**Table 3.** Jumper knobs and what each demonstrates.

| Knob | Setting | Demonstrates |
|---|---|---|
| CM1 + CM2 | Fitted = CM choke L1 bypassed | Effect of the common-mode filter |
| YCAP | Fitted = input Y-caps C2/C3 (2 × 10 nF) to chassis | Y-capacitor return path for CM current |
| DM | Fitted = DM inductor L2 bypassed | Effect of the differential-mode filter |
| RG | 1-2 = 4.7 Ω, 3-4 = 22 Ω, 5-6 = 100 Ω (fit exactly one) | Gate resistance → dv/dt → spectrum |
| CASE | Open = floating, 1-2 = PGND, 3-4 = CHASSIS, 5-6 = C14 1 nF to PGND | Potential of the aluminium case = Q1 heatsink (Williams cases, "don't float the heatsink"; lecture figure heatsink/housing) |
| SNUB | Fitted = 100 pF + 47 Ω drain snubber | Ringing and dv/dt reduction |
| SGCH | Fitted = secondary 0 V earthed to chassis | Earthed output: pri-sec CM current flows through the chassis |
| YPS | 1-2 = C23 1 nF, 3-4 = C24 10 nF primary-to-secondary | External Y-cap across the barrier (Ex. 11.2) |
| MH5 / MH1–MH4 | MH5: metal stand-off from the case to the CASE jumper; MH1–MH4: nylon | The case is connected only through the CASE jumper (rubber feet keep it off the plate) |

## 2.3 Specifications

**Table 4.** Design specification.

| Parameter | Specification | Notes |
|---|---|---|
| Input voltage | 24 V nominal, 20–30 V | Lab bench supply through 2 × 5 µH LISNs |
| Output | 12.2 V, 1 A (12.2 W) | 4 mm sockets J3/J4; 1 kΩ preload R13 |
| Isolation | Functional (no safety rating needed at 24 V) | T1 rated 1500 $V_{AC}$; 3 mm PGND–SGND gap on the PCB |
| Topology / control | Flyback, DCM, peak-current mode | UC3845B + TL431 + LTV-817S |
| Switching frequency | ≈ 150 kHz | 153.6 kHz from RT/CT; inside the CISPR band B start (150 kHz) |
| Efficiency | ≥ 80 % at 1 A, Rg = 4.7 Ω | Lower at 100 Ω by design (slow edge) |
| Output ripple / regulation | ≤ 100 mV p-p; ±1 % over load and line | |
| Protection | PTC fuse 2 A, reverse-polarity diode, cycle-by-cycle current limit | No soft-start (UC3845) |
| Thermal | All parts below rating at 25 °C ambient, continuous operation | Including the 100 Ω setting |
| Mechanical | 2-layer FR-4 1.6 mm, 100 × 76 mm, in a 112 × 94 × 33 mm aluminium U-channel case (3 mm) with a transparent 3 mm PMMA cover, Q1 on the case floor | Case on rubber feet on the aluminium reference plate (chassis) |
| Manufacturing | JLCPCB SMT assembly of the top side; magnetics, Q1, sockets and headers by hand; case bent in the workshop | LCSC part numbers checked 2 Oct 2026 |

# 3 Circuit topology

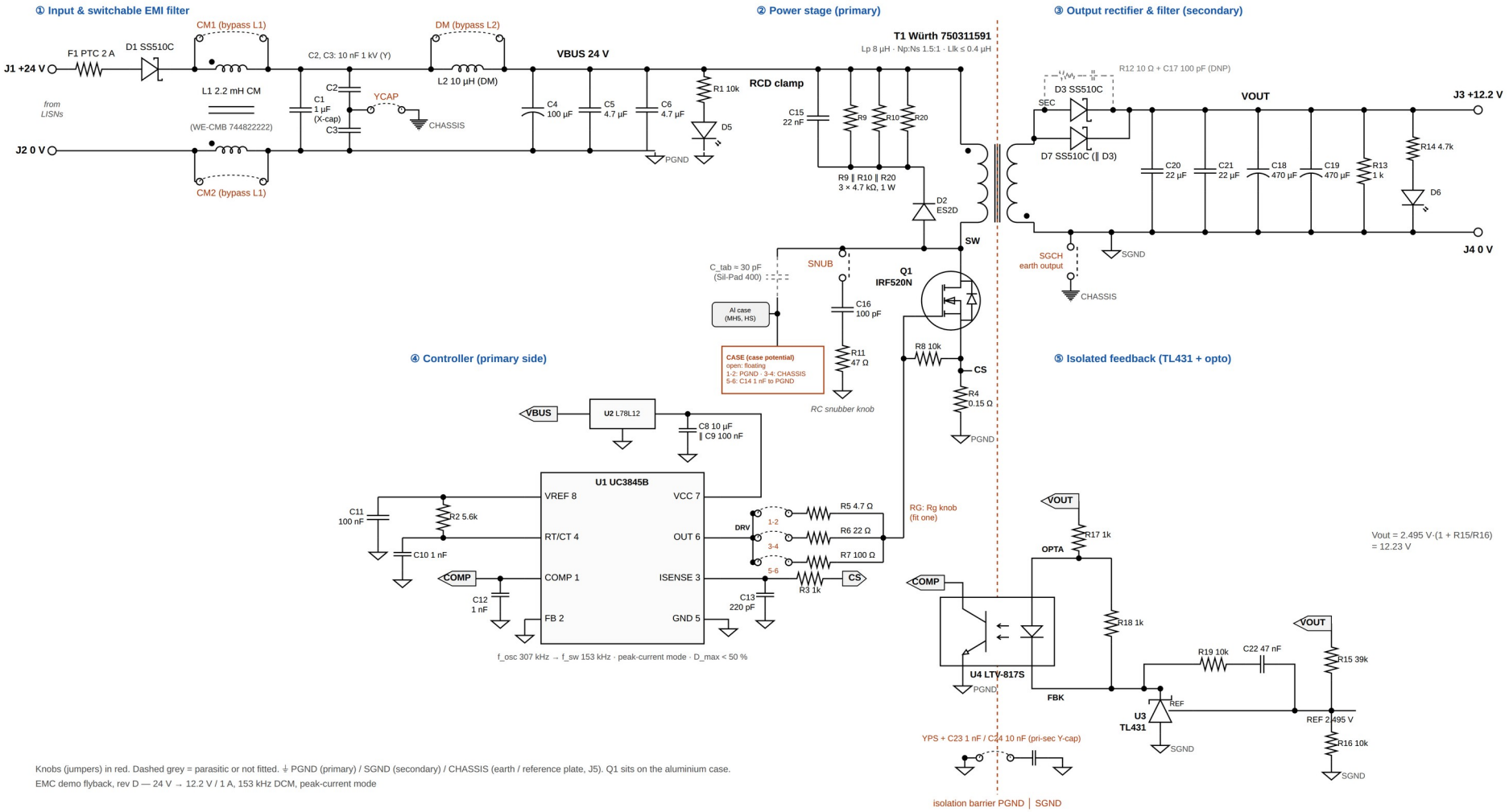



**Figure 1.** Complete circuit topology (drawn in draw.io, file EMC_flyback_topology.drawio). ① switchable input EMI filter, ② primary power stage with RCD clamp, RC-snubber and case (heatsink) knobs, ③ secondary rectifier and output filter, ④ UC3845B controller with the Rg knob, ⑤ isolated TL431/opto feedback. The dashed red line is the isolation barrier; jumper knobs are shown in red, parasitic or not-fitted parts in dashed grey.

Figure 1 shows the complete circuit. The blocks are:

- ① **Input and switchable EMI filter.** PTC fuse F1 (2 A) and series Schottky D1 (reverse-polarity protection), common-mode choke L1 (2.2 mH, bypass CM1/CM2), X-capacitor C1 (1 µF), Y-capacitors C2/C3 (10 nF, 1 kV) whose midpoint is connected to the chassis with YCAP, differential-mode inductor L2 (10 µH, bypass DM) and the bulk/hot-loop capacitors C4 (100 µF) and C5/C6 (4.7 µF). Downstream of L1 the negative rail is the primary ground PGND.
- ② **Primary power stage.** T1 primary (dot at VBUS), MOSFET Q1 (IRF520N) on the aluminium case, current-sense resistor R4 (0.15 Ω). The RCD clamp D2/C15/R9‖R10‖R20 (3 × 4.7 kΩ) limits the drain voltage after turn-off. SNUB connects the RC snubber C16/R11 from the drain to PGND. The TO-220 tab (drain) couples to the case through the Sil-Pad capacitance $C_{tab}$ ≈ 30 pF; CASE selects the case potential (through the metal stand-off MH5).
- ③ **Secondary.** T1 secondary (dot at SGND), two Schottky rectifiers D3 ‖ D7 (SS510C), ceramic C20/C21 (2 × 22 µF) for the high-frequency current and electrolytic C18/C19 (2 × 470 µF) for bulk storage, preload R13 and LED D6. An RC snubber R12/C17 across D3 is provided but not fitted. SGCH earths the output; YPS adds a primary-to-secondary Y-capacitor.
- ④ **Controller.** UC3845B, supplied with 12 V from the L78L12 linear regulator U2 (no auxiliary winding is used, so the controller runs as soon as VBUS is present). RT/CT = R2/C10 set the oscillator; the 1 V current-sense input is filtered by R3/C13. The output drives the gate through one of the three resistors R5/R6/R7 selected by RG; R8 holds the gate low.
- ⑤ **Isolated feedback.** The TL431 compares the divided output voltage (R15/R16) with 2.495 V. Its cathode current flows through the opto LED (series resistor R17, pre-bias R18). The opto transistor pulls the UC3845 COMP pin down; FB is grounded, so the internal error amplifier only acts as a 0.8 mA pull-up on COMP. R19/C22 compensate the TL431.

Three grounds are used: PGND (primary), SGND (secondary) and CHASSIS (earth, i.e. the aluminium reference plate, connected through J5). The aluminium case is a separate node (HS) that CASE connects to one of them or leaves floating. Only T1, the opto U4, the optional YPS capacitors and the chassis track cross the 3 mm gap between the PGND and SGND copper pours.

# 4 Parameter design

The values below come from design_calc.py (in the simulation folder of the report). Where a quantity was also simulated, the simulated value is given for comparison; Section 5.7 collects all comparisons.

## 4.1 Switching frequency and controller

The UC3845B oscillator is set by RT = R2 = 5.6 kΩ and CT = C10 = 1 nF (C0G). The UC3845 has an internal toggle flip-flop, so the switching frequency is half the oscillator frequency and the maximum duty cycle is just below 50 %:

$$f_{osc} = \frac{1.72}{R_T C_T} = 307\,kHz\,,\; f_{sw} = \frac{f_{osc}}{2} = 153.6\,kHz \tag{1}$$

153.6 kHz puts the fundamental just above the 150 kHz start of CISPR band B, so the harmonics are clearly visible on the LISN spectrum. The simulated switching frequency is 152.5 kHz (the model uses a 2.8 V/1.2 V oscillator window).

## 4.2 Operating mode and transformer

The transformer is the off-the-shelf Würth 750311591 (EFD25 SMD): $L_p$ = 8 µH, $N_p$:$N_s$:$N_a$ = 1.5:1:0.67, leakage inductance ≤ 0.4 µH (k ≈ 0.975), $I_{sat}$ = 20 A, 1500 $V_{AC}$ isolation. The auxiliary winding is not used. With n = $N_p/N_s$ = 1.5 the reflected voltage is

$$V_{OR} = n\left(V_o + V_F\right) = 1.5 \cdot \left(12.23 + 0.55\right) = 19.2\,V \tag{2}$$

In DCM all energy stored in $L_p$ during the on-time is transferred to the output, so the peak current depends only on the power through the transformer ($P_{Lp}$ ≈ $P_o$ + diode + copper losses ≈ 12.9 W) and not on the input voltage:

$$I_{pk}=\sqrt{\frac{2P_{Lp}}{L_p f_{sw}}}=4.59\,A,t_{on}=\frac{L_p I_{pk}}{V_{in}},t_{dem}=\frac{L_p I_{pk}}{V_{OR}} \tag{3}$$

The converter stays in DCM if $L_p$ is below the boundary inductance at minimum input voltage. With $D_b = V_{OR}/(V_{in,min} + V_{OR}) = 0.49$:

$$L_{crit}=\frac{\left(V_{in,min}D_b\right)^2}{2P_{Lp}f_{sw}}=24.1\,\mu H \gg L_p=8\,\mu H \tag{4}$$

**Table 5.** Operating point over the input range (calculated).

| $V_{in}$ | Duty D | $t_{on}$ | $t_{dem}$ | Idle (ringing) time | Comment |
|---|---|---|---|---|---|
| 20 V | 0.282 | 1.83 µs | 1.91 µs | 2.76 µs | Largest duty, still < 0.48 |
| 24 V | 0.235 | 1.53 µs | 1.91 µs | 3.07 µs | Simulated D = 0.244 |
| 30 V | 0.188 | 1.22 µs | 1.91 µs | 3.37 µs | Most ringing time |

The large idle time (≈ 3 µs, almost half the period) makes the converter deep DCM over the full range: there is no right-half-plane zero in the control loop, no sub-harmonic oscillation and no reverse recovery in D3. The price is a high peak-to-average current ratio. The RMS currents follow from the triangular waveforms:

$$I_{p,rms}=I_{pk}\sqrt{\frac{D}{3}}=1.28\,A,I_{s,rms}=n\cdot I_{pk}\sqrt{\frac{D_s}{3}}=2.15\,A,I_{s,pk}=n\cdot I_{pk}=6.9\,A \tag{5}$$

Simulation gives $I_{pk}$ = 4.73 A, $I_{p,rms}$ = 1.43 A, $I_{s,rms}$ ≈ 2.12 A and $I_{s,pk}$ = 8.3 A, shared equally by D3 and D7 (4.1 A each). The secondary peak is higher than $n{\cdot}I_{pk}$ because the leakage inductance delays the current transfer, which causes an overshoot followed by ringing (Figure 4).

## 4.3 Semiconductor stresses and selection

The drain voltage is the input voltage plus the clamp voltage $V_C$. $V_C$ is chosen at about $2.2{\cdot}V_{OR}$ ≈ 42 V, which balances clamp loss (lower for high $V_C$) against MOSFET voltage margin. The output diode sees the output voltage plus the input voltage reflected to the secondary:

$$V_{DS,max}=V_{in}+V_C\approx 24+42=66\,V,V_{R,D3}=V_o+\frac{V_{in,max}}{n}=32\,V \tag{6}$$

**Table 6.** Component stresses (24 V, 1 A unless stated) and ratings.

| Part | Stress (calc. / sim.) | Rating | Margin / comment |
|---|---|---|---|
| Q1 IRF520N (TO-220) | $V_{DS}$ 66 / 66.4 V; ~73 V at 30 V<br>$I_{D,pk}$ 4.6 / 4.7 A (8.5 A cold start) | 100 V, 9.7 A, $R_{DS(on)}$ 0.2 Ω | 27 % voltage margin at 30 V; low $Q_g$ (25 nC) for a 1 A driver |
| D3, D7 SS510C (SMC, in parallel) | $V_R$ 32 V + ringing (38 V sim.)<br>per diode $I_{F,avg}$ 0.5 A, $I_{F,pk}$ 4.1 A | 100 V, 5 A each | Schottky: no reverse recovery; 0.17 W each |
| D2 ES2D (SMA) | $V_R \approx V_C$ = 42–67 V; pulses at turn-off | 200 V, 2 A, $t_{rr}$ 20 ns | Fast recovery keeps the clamp efficient |
| D1 SS510C | 0.62 A average, 0.3 W | 100 V, 5 A | Reverse-polarity protection |
| T1 750311591 | $I_{pk}$ 4.7 A (8.5 A transient) | $I_{sat}$ 20 A | Factor 2.4 even at a cold start |
| R4 0.15 Ω (2512) | 0.25 / 0.30 W | 1 W | Current limit 1 V/0.15 Ω = 6.7 A |
| R9, R10, R20 4.7 kΩ (2512) | 0.30 W each (sim.); hot spot ≈ 68 °C | 1 W each (70 °C, derated to 155 °C) | Three parts on local copper + thermal vias (rev C) |
| C15 22 nF 100 V X7R | ≈ 42 V DC + 8 V ripple | 100 V | |
| C18/C19 470 µF 25 V | 12.2 V; ripple current ≈ 1.9 $A_{rms}$ total (shared with C20/C21) | 25 V | **Check** the ripple-current rating of RVT1E471M1010 |
| U2 L78L12 (SOT-89) | 0.27 W at 30 V input | ~0.5 W (SOT-89, copper) | Supplies UC3845 + gate charge |

## 4.4 RCD clamp

At turn-off the energy stored in the leakage inductance ($L_{lk} = L_p(1-k^2)$ = 0.40 µH) is dumped into C15 through D2. Because the clamp also takes part of the magnetising current while the secondary current builds up, the clamp loss is larger than the leakage energy alone:

$$P_{lk} = \frac{1}{2} L_{lk} I_{pk}^2 f_{sw} = 0.65\,W\,,\; P_{clamp} = P_{lk} \frac{V_C}{V_C - V_{OR}} = 1.19\,W \tag{7}$$

$$R_{clamp} = \frac{V_C^2}{P_{clamp}} = 1485\,\Omega \rightarrow R9 \parallel R10 \parallel R20 = 3 \times 4.7\,k\Omega = 1.57\,k\Omega \tag{8}$$

C15 = 22 nF gives an RC time constant of 33 µs (5 switching periods) and a ripple of $V_C/(R\,C\,f_{sw})$ ≈ 8 V. Rev B used 2 × 3.0 kΩ (1.5 kΩ, 0.48 W per resistor). Rev C splits the resistor into three 4.7 kΩ / 1 W parts (1.57 kΩ, 4 % higher), which raises the clamp voltage slightly and lowers the clamp loss. The closed-loop simulation gives a drain peak of 66.4 V, an average clamp voltage of 38 V above VBUS and a clamp loss of 0.91 W, i.e. 0.30 W per resistor.

## 4.5 RC snubber knob and secondary snubber

After the clamp diode stops conducting, $L_{lk}$ rings with the drain capacitance ($C_{oss}$ of the IRF520N, the winding capacitance and the layout, together ≈ 150 pF):

$$f_{ring} = \frac{1}{2\pi\sqrt{L_{lk} C_{par}}} \approx 21\,MHz\,,\; Z_0 = \sqrt{\frac{L_{lk}}{C_{par}}} \approx 52\,\Omega \tag{9}$$

The snubber resistor is matched to the characteristic impedance (R11 = 47 Ω) and the snubber capacitor is of the same order as $C_{par}$ (C16 = 100 pF C0G, 2 kV). The loss is at most $C16 \cdot V_{DS}^2 \cdot f_{sw}$ ≈ 0.07 W (simulated 0.065 W). The secondary snubber R12 = 10 Ω / C17 = 100 pF across D3 is laid out but not fitted, as a reserve if the D3 ringing turns out to be dominant.

## 4.6 Gate drive and the Rg knob

The UC3845 totem-pole output (≈ 1 A peak, $V_{OH}$ ≈ VCC − 1.5 V = 10.5 V) drives the gate through one of three resistors. The gate charge loss is $Q_g \cdot VCC \cdot f_{sw}$ ≈ 0.05 W. The gate resistance sets the Miller-plateau current and

therefore the drain dv/dt. That determines the second corner frequency of the trapezoidal spectrum, $1/(\pi\, t_r)$, and the common-mode current injected through the tab capacitance:

$$I_{CM} = C_{tab}\frac{dv}{dt}, f_{c2} = \frac{1}{\pi t_r} \tag{10}$$

**Table 7.** Rg knob: switching speed versus loss (dv/dt from simulation, $t_r$ = full 66 V swing at that dv/dt).

| Rg | dv/dt | $t_r$ | $f_{c2}$ | $I_{CM}$ via 30 pF | Q1 loss (sim.) | η (sim.) | $T_j$ (thermal model) |
|---|---|---|---|---|---|---|---|
| 4.7 Ω (R5) | 5.0 V/ns | 13 ns | 24 MHz | 150 mA | 0.73 W | 83.5 % | 41 °C |
| 22 Ω (R6) | 2.5 V/ns | 26 ns | 12 MHz | 75 mA | 0.97 W | 82.2 % | 45 °C |
| 100 Ω (R7) | 0.65 V/ns | 101 ns | 3.2 MHz | 20 mA | 1.92 W | 77.3 % | 58 °C |

$T_j$ comes from the board thermal model of Section 5.8 (25 °C ambient): $R_{th,jc}$ = 3.1 K/W (IRF520N), ≈ 6 K/W for the Sil-Pad 400 (Bergquist TO-220 data: 4.6–8.5 K/W depending on the mounting pressure), 0.5 K/W spreading into the 3 mm floor, and the case itself (≈ 8 K/W to ambient, bare aluminium, at these small temperature rises). Even the 100 Ω setting can therefore run continuously (case ≈ 40 °C).

## 4.7 Current sensing and protection

R4 = 0.15 Ω converts the primary current into the ISENSE signal. The UC3845 clamps the current-sense threshold at 1 V, which gives a nominal cycle-by-cycle limit of 6.7 A, 40 % above the steady-state peak. At full load the COMP voltage is 3.15 V, giving a threshold of ($V_{COMP}$ − 1.4 V)/3 = 0.58 V. R3 = 1 kΩ and C13 = 220 pF (τ = 220 ns, 3.4 % of the period) suppress the leading-edge spike caused by the discharge of the drain capacitance at turn-on. Together with the comparator and turn-off delays they let the current overshoot the threshold by

$$\Delta I \approx \frac{V_{in}}{L_p}\left(\tau_{RC} + t_{d,comp} + t_{d,off}\right) \approx 3\,A/\mu s \cdot 0.32\,\mu s \approx 0.96\,A \tag{11}$$

so at the current limit the real peak is about 7.6 A, which matches the 7.5 A seen in the cold-start simulation (Figure 7). Further protection: the PTC F1 (2 A hold, 0.62 A nominal input current) and the series diode D1 against reverse polarity. There is no over-voltage protection; if the opto fails open, the converter runs at the current limit and the output rises until C18/C19 (25 V) are at risk. That is acceptable for a supervised lab demonstrator.

## 4.8 Output filter

The output capacitor bank must carry the secondary ripple current and keep the switching ripple low:

$$I_{C,rms} = \sqrt{I_{s,rms}^2 - I_o^2} = 1.91\,A \tag{12}$$

The two 22 µF X5R ceramics (C20/C21, ≈ 3 mΩ each) are placed directly at D3/D7 and take most of the high-frequency current; the two 470 µF electrolytics (C18/C19, ≈ 60 mΩ each) provide the bulk capacitance for the load steps. An upper bound for the ripple is $I_{s,pk}$·$ESR_{el}$ ≈ 0.21 V; with the ceramics in parallel the simulation gives 75 mV peak-to-peak. The output pole of the DCM flyback is

$$f_p = \frac{2}{2\pi R_L C_o} = 26.5\,Hz, f_{z,ESR} = \frac{1}{2\pi ESR \cdot C_{el}} \approx 5.6\,kHz \tag{13}$$

## 4.9 Input capacitors and EMI filter

**Input capacitors.** The primary current pulses ($I_{p,rms}$ ≈ 1.4 A, of which 0.62 A DC) cause ≈ 1.1 $A_{rms}$ ripple current. It is carried by the two 4.7 µF X7R ceramics C5/C6, which form the primary hot loop with T1, Q1 and R4. C4 (100 µF electrolytic) provides low-frequency storage and damping.

**Differential-mode filter.** L2 (10 µH) and C4 + C5 + C6 (≈ 109 µF) form a second-order low-pass filter:

$$f_{0,DM} = \frac{1}{2\pi\sqrt{L_2 C_{in}}} = 4.8\,kHz,\; Z_0 = \sqrt{\frac{L_2}{C_{in}}} = 0.30\,\Omega \tag{14}$$

The ideal attenuation at $f_{sw}$ is about 60 dB. The ESR of C4 (≈ 0.3 Ω) is equal to $Z_0$, so the filter resonance is well damped (Q ≈ 1). The converter's negative input impedance $|V_{in}^2/P_{in}|$ ≈ 39 Ω is two orders of magnitude above $Z_0$, so the Middlebrook stability criterion is met with a large margin. The leakage inductance of L1 adds to L2 for DM noise.

**Common-mode filter.** L1 (Würth WE-CMB 744822222, 2.2 mH, 2 A) with C2 + C3 (20 nF) to chassis (YCAP fitted) gives a CM corner frequency of

$$f_{0,CM} = \frac{1}{2\pi\sqrt{L_1(C_2 + C_3)}} = 24\,kHz \tag{15}$$

Without YCAP the Y-capacitors have no return path, so L1 alone (against the LISN impedance) provides CM attenuation. This is one of the demonstrations. No copper pour is placed on the LISN side of L1, so the pour cannot bypass the choke.

## 4.10 Feedback and compensation

### *Set point and bias*

$$V_o = V_{ref}\left(1 + \frac{R_{15}}{R_{16}}\right) = 2.495 \cdot \left(1 + \frac{39\,k}{10\,k}\right) = 12.23\,V \tag{16}$$

With FB grounded, the UC3845 error amplifier sources 0.8 mA (typical) into COMP. In steady state the opto transistor sinks this current minus the load of the internal R/2R divider, so at CTR = 1 the LED current is ≈ 0.74 mA. R18 = 1 kΩ across the LED pre-biases the TL431 with 1.05 mA, so its cathode current (≈ 1.8 mA) is always above the 1 mA minimum, even when the LED is almost off. The TL431 cathode sits at ≈ 9.4 V, far above its 2.5 V minimum. R17 = 1 kΩ limits the LED current to < 9 mA in a transient.

### *Small-signal model*

For DCM peak-current mode the output voltage is proportional to the peak current ($V_o = I_{pk}\cdot(L_p f_{sw} R_L/2)^{1/2}$), so the control-to-output gain is a single pole with the ESR zero:

$$G_{vc}(s) = \frac{1}{3\,R_4}\,\frac{V_o}{I_{pk}} \cdot \frac{1 + s/\omega_z}{1 + s/\omega_p},\; G_{vc0} = 5.92\,(15.4\,dB) \tag{17}$$

The feedback path consists of the TL431 type-2 integrator (zero at 1/(2πR19C22) = 339 Hz), the fast lane through R17, the opto (CTR, pole $f_{opto}$ ≈ 20 kHz) and the COMP node (C12 with the internal divider load $R_{div}$):

$$H(s) = \frac{CTR \cdot Z_{comp}}{R_{17} + r_d}\left(1 + \frac{R_{19} + 1/(sC_{22})}{R_{15}}\right)\frac{1}{1 + s/\omega_{opto}},\; Z_{comp} = R_{div} \parallel \frac{1}{sC_{12}} \tag{18}$$

Here $r_d$ is the dynamic resistance of the LED (≈ 60 Ω at 0.74 mA). The loop gain $T = G_{vc}\cdot H$ is plotted in Figure 2 for CTR = 0.5, 1 and 2. The LTV-817S without a rank letter spreads from 0.5 to 6. Above ≈ 1 kHz the fast lane dominates and the integrator only trims the DC error, which explains the slow tail after a load step (Section 5.5).

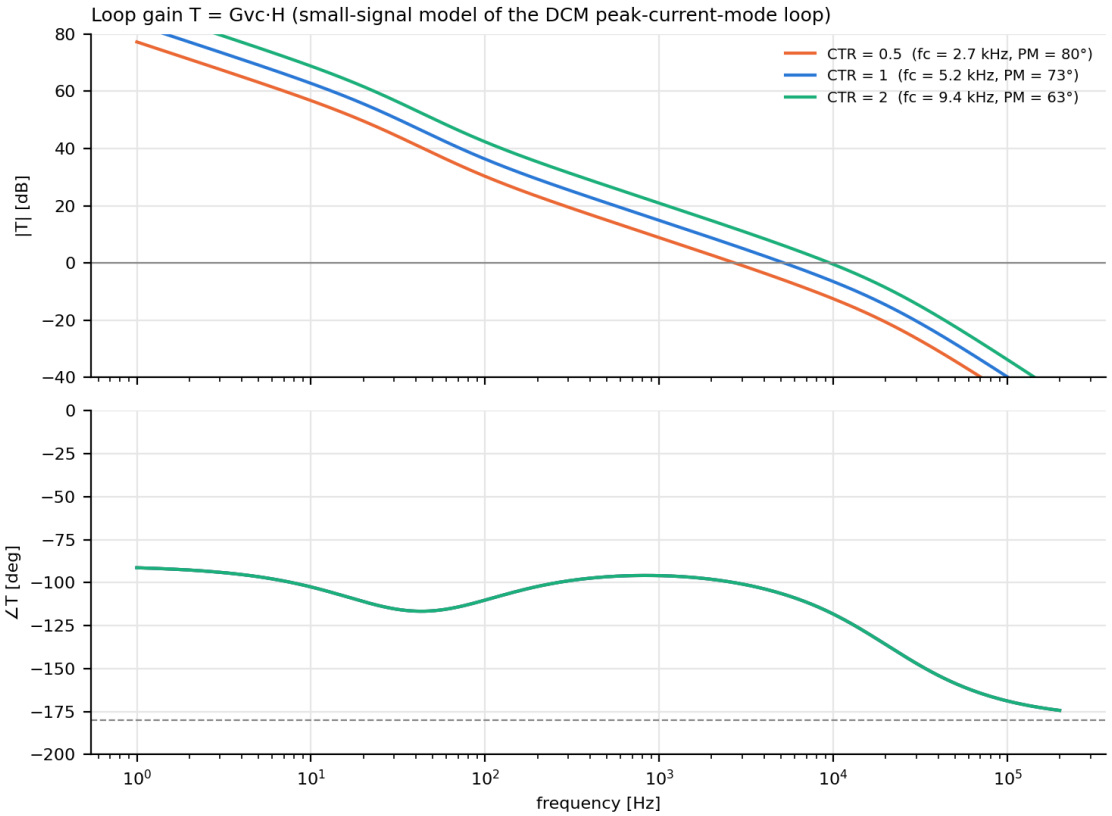



**Figure 2.** Loop gain from the small-signal model ($R_{div}$ = 30 kΩ assumed for the UC3845 internal COMP load). The phase does not depend on CTR, so the three phase curves coincide.

**Table 8.** Crossover frequency and phase margin.

| CTR | $f_c$ | Phase margin | Comment |
|---|---|---|---|
| 0.5 | 2.7 kHz | 80° | Slowest response |
| 1 (default) | 5.2 kHz | 73° | Step response time constant ≈ $1/(2\pi f_c)$ ≈ 30 µs, consistent with the simulated ≈ 40 µs |
| 2 | 9.4 kHz | 63° | Still ≈ $f_{sw}/16$ |

## 4.11 Auxiliary supply and indicators

U2 (L78L12) feeds VCC from VBUS: about 11 mA for the UC3845 plus 4 mA gate charge, i.e. 0.18 W at 24 V and 0.27 W at 30 V. VCC is decoupled with C8 (10 µF) and C9 (100 nF), the input with C7 (100 nF) and VREF with C11 (100 nF). The red LED D5 (R1 10 kΩ, 2.2 mA) shows VBUS, the green LED D6 (R14 4.7 kΩ, 2.2 mA) shows VOUT. R13 (1 kΩ, 0.15 W in 1206) is a 12 mA preload that keeps the converter switching regularly at no load.

## 4.12 Losses and thermal design

Figure 3 breaks down the simulated losses at 24 V and 1 A. The RCD clamp (0.91 W) is the largest single loss for the fast settings; Q1 takes over at 100 Ω. The two parallel rectifiers lose 0.35 W together, 0.06 W less than a single SS510C because each runs at half the current. The RC snubber slightly reduces the Q1 turn-off loss (it slows the dv/dt), so the efficiency with the snubber is marginally higher (83.7 %). These are power-stage losses only: the simulation feeds VBUS directly and supplies VCC from an ideal source. On the real board the reverse-polarity diode D1 (≈ 0.26 W), the L78L12 path (≈ 0.36 W from VBUS), the preload R13 and LED D6 (≈ 0.18 W extra output) and LED D5 (0.05 W) add ≈ 0.9 W at the input, so the board-level efficiency at 1 A is ≈ 78–79 %. Section 5.8 turns these losses into component temperatures.

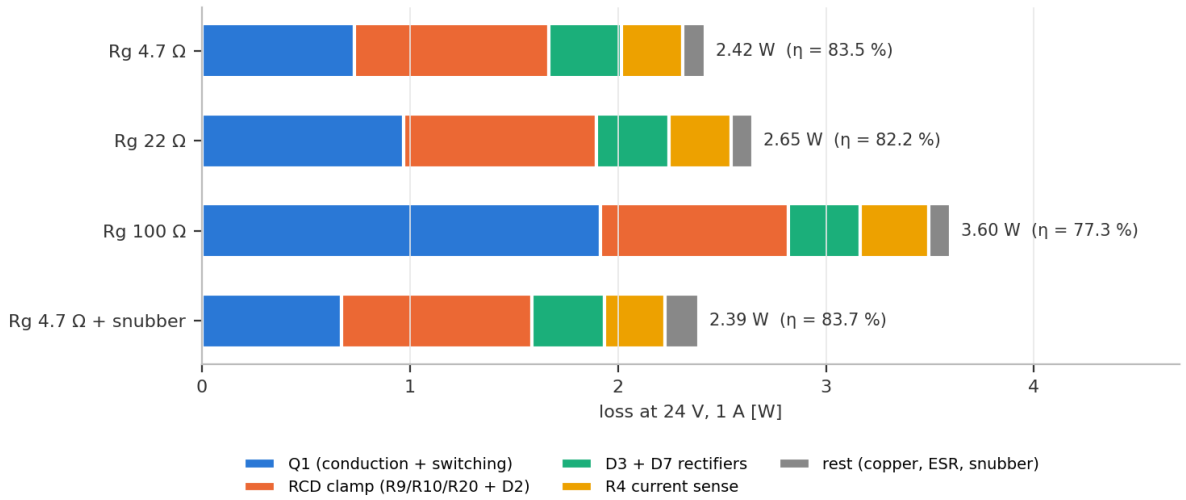



**Figure 3.** Simulated loss breakdown and efficiency at 24 V, 1 A (closed loop).

# 5 Simulation

## 5.1 Models and set-up

Two LTspice schematics were made, both generated from the board values:

- Both include the rev C changes: the clamp resistor as R9 ‖ R10 ‖ R20 (3 × 4.7 kΩ) and the rectifier as D3 ‖ D7. Rev D changes only the mechanics and rev E only the placement of the power cell, so the simulations are unchanged; the case capacitance is treated separately in Section 6.3. The 10 nH hot-loop stray inductance of the models is of the same order as the rev E layout estimate (14–18 nH, Section 6.1).
- **EMC_flyback_power_stage.asc**: open-loop power stage with a fixed duty cycle (0.245), used for stresses and switching speed.
- **EMC_flyback_closed_loop.asc**: the same power stage plus the real control loop, using behavioural models of the UC3845B, TL431 and LTV-817S (EMC_flyback_control.lib). These were written from datasheet typical values: oscillator 2.8 V/1.2 V with 8.3 mA discharge, toggle flip-flop, ($V_{COMP}$ − 1.4 V)/3 threshold with a 1 V clamp, ≈ 70 ns current-sense delay, 0.8 mA COMP pull-up, TL431 with 60 dB gain and 1.5 kHz pole, and an opto with CTR and a 20 kHz pole.

Q1 uses a VDMOS model fitted to the IRF520N datasheet ($R_{DS(on)}$ ≈ 0.16 Ω at 10 V, $Q_g$ ≈ 25 nC). The diodes use generic models with the datasheet $V_F$, $C_j$ and $t_{rr}$. T1 is modelled as coupled inductors ($L_p$ = 8 µH, k = 0.9747, winding resistances 15/10 mΩ), with a 10 nH hot-loop stray inductance. The secondary ground is tied to the simulator ground, so common-mode paths are not modelled. Because LTspice was not available, both schematics were converted (from their geometry) into ngspice netlists and run there: 2–5 ns maximum time step, Gear integration.

## 5.2 Steady-state waveforms

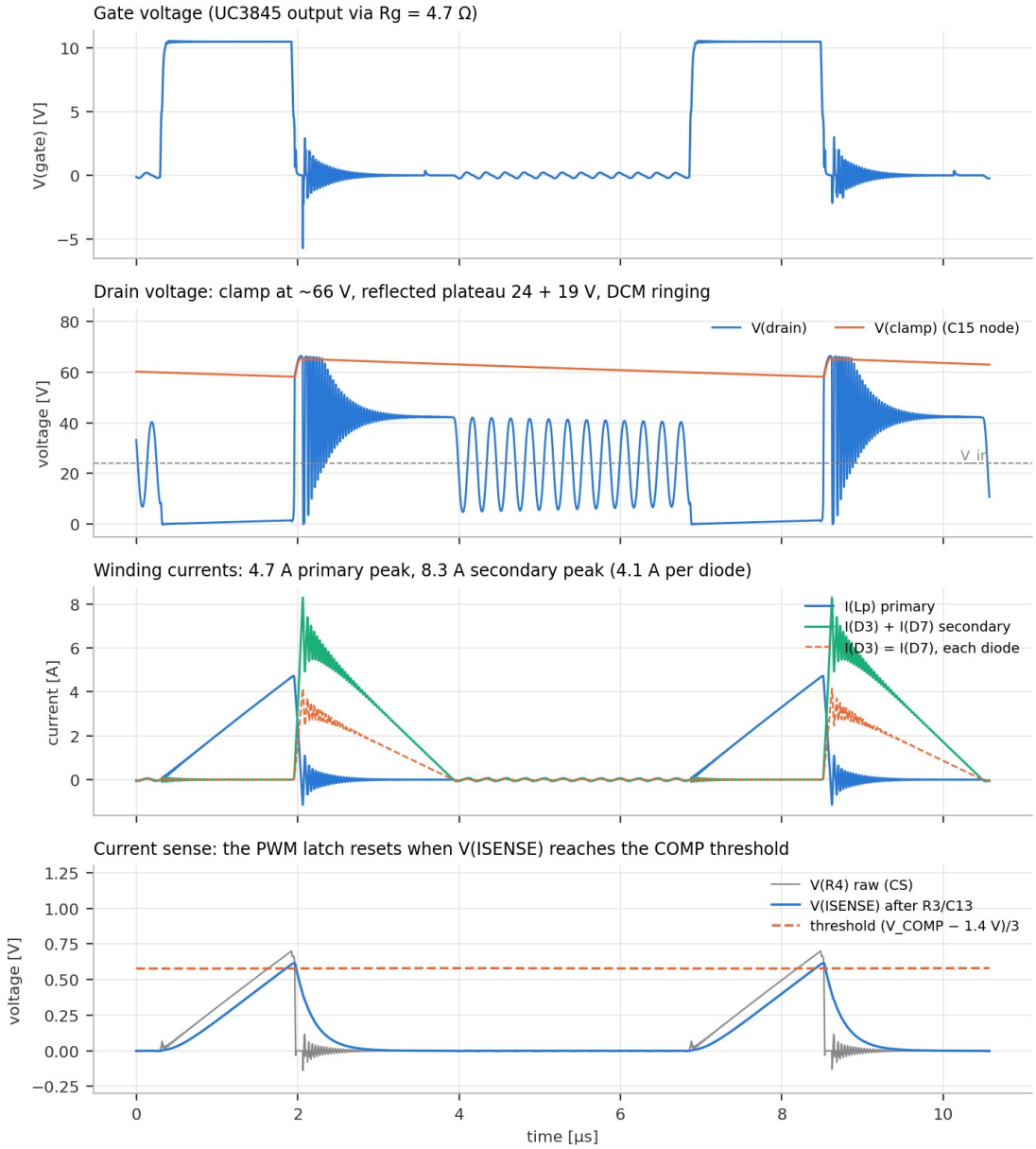



**Figure 4.** Steady-state waveforms at 24 V, 1 A, Rg = 4.7 Ω (closed loop): gate, drain and clamp voltage, primary and secondary current, current-sense signals.

The waveforms show the expected DCM operation: the primary current ramps to 4.8 A in 1.6 µs. At turn-off the drain is clamped at ≈ 66 V and then settles at $V_{in}$ + $V_{OR}$ ≈ 43 V while the secondary conducts (≈ 2 µs). After the secondary current reaches zero, the drain rings around $V_{in}$ with an amplitude of ≈ $V_{OR}$ for the rest of the period. The next turn-on happens at a random point of this ring, which spreads the turn-on loss and the turn-on spike. The R3/C13 filter delays ISENSE by ≈ 0.2 µs; the latch resets when ISENSE reaches the 0.58 V threshold.

## 5.3 Rg knob and RC snubber: edges and spectrum

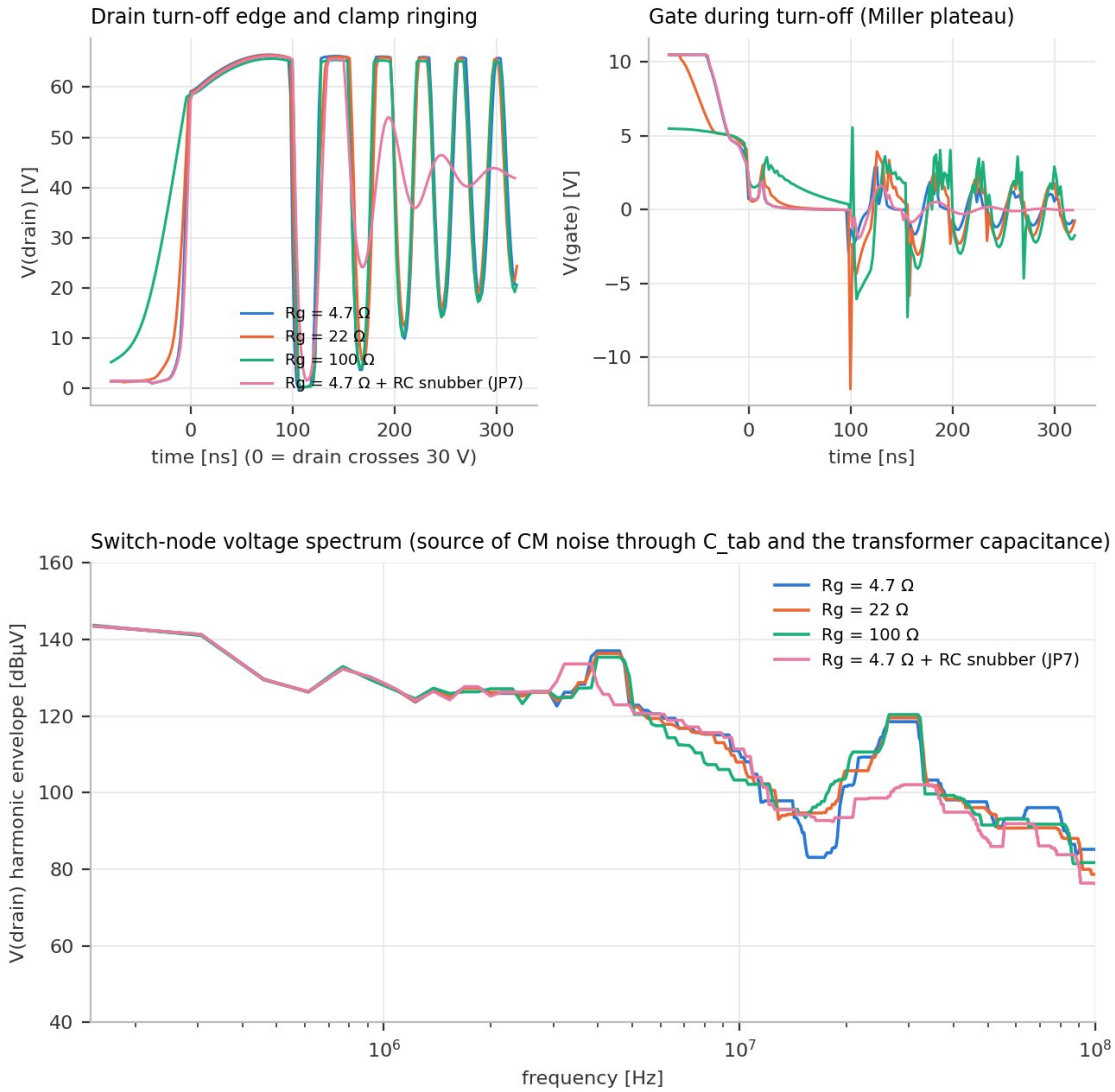



**Figure 5.** Drain turn-off edge, gate voltage and harmonic envelope of the drain voltage for the three gate resistors and with the RC snubber (24 V, 1 A, closed loop).

The gate resistor slows the first part of the drain edge (the Miller plateau), but in this model the edge is then completed by the leakage inductance, and the clamp ringing at ≈ 16 MHz (66 V ↔ ≈ 5 V, about 200 ns) is almost independent of Rg. In the switch-node spectrum the 100 Ω setting lowers the 5–15 MHz band by 5–8 dB, while the 20–30 MHz peak from the clamp ringing stays. The RC snubber damps that ringing within one cycle and lowers the 20–30 MHz peak by ≈ 20 dB, at the cost of a new, lower peak near 3.5 MHz.

**Consequence for the lecture.** The simulation suggests that the snubber knob (SNUB) will show a clearer effect above 20 MHz than the Rg knob, and that the Rg effect will be most visible in the 3–15 MHz range. The real ringing depends on layout parasitics that the model does not contain, so this is a prediction to be checked with the LISN measurement, not a conclusion.

## 5.4 Losses and efficiency

**Table 9.** Simulated power balance at 24 V, 1 A (closed loop).

| Case | $P_{in}$ | $P_{out}$ | Q1 | Clamp | D3 + D7 | R4 | η |
|---|---|---|---|---|---|---|---|
| Rg 4.7 Ω | 14.66 W | 12.24 W | 0.73 W | 0.93 W | 0.35 W | 0.29 W | 83.5 % |
| Rg 22 Ω | 14.89 W | 12.24 W | 0.97 W | 0.93 W | 0.35 W | 0.30 W | 82.2 % |
| Rg 100 Ω | 15.84 W | 12.24 W | 1.92 W | 0.90 W | 0.35 W | 0.33 W | 77.3 % |
| Rg 4.7 Ω + snubber | 14.63 W | 12.24 W | 0.67 W | 0.91 W | 0.35 W | 0.29 W | 83.7 % |

## 5.5 Closed-loop regulation and load steps

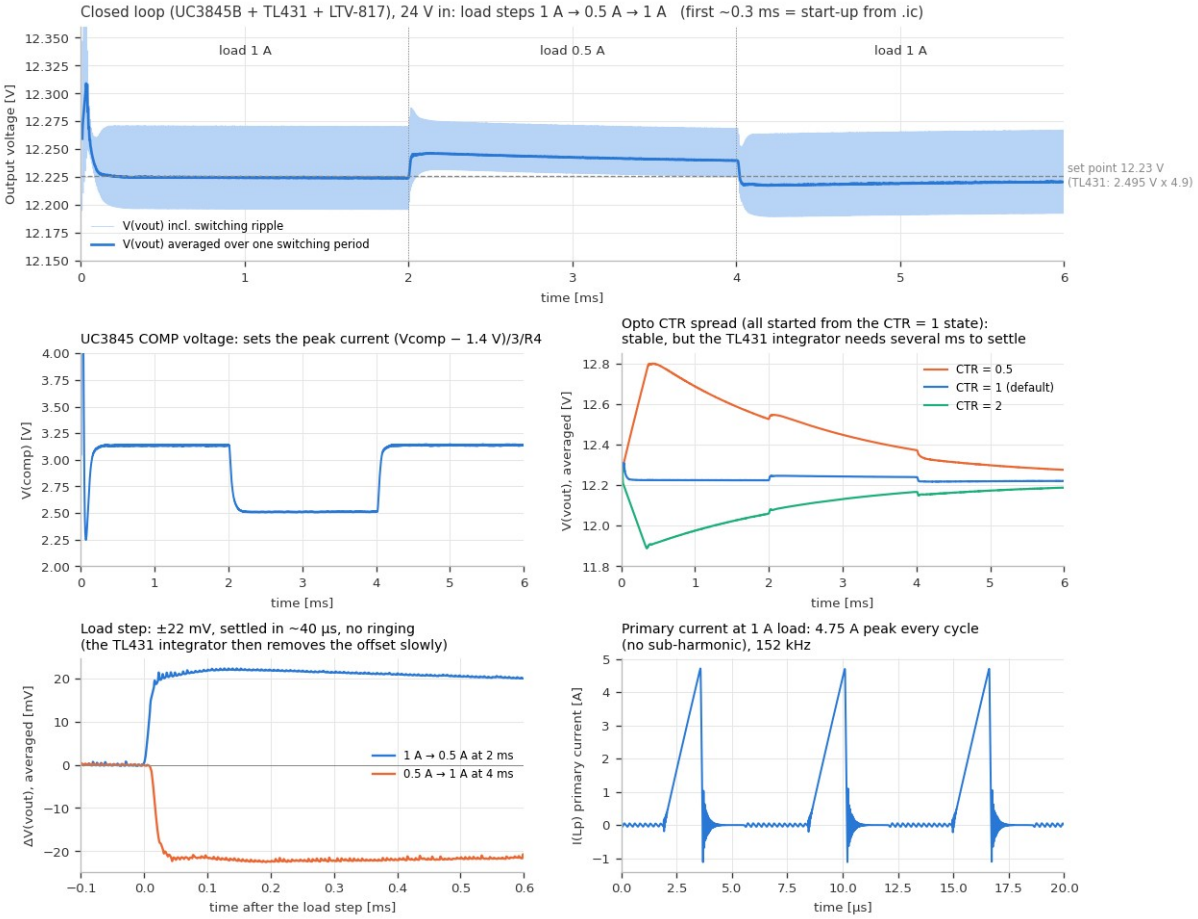



**Figure 6.** Closed-loop simulation: output voltage with load steps 1 A → 0.5 A → 1 A, COMP voltage, opto CTR spread, step detail and primary current.

The output is regulated at 12.22–12.24 V. A 0.5 A load step gives a ±22 mV deviation (switching-period average) that settles within ≈ 40 µs without ringing. The remaining ≈ 20 mV offset is then removed slowly by the TL431 integrator (time constant of a few ms, set by R15·C22). With CTR = 0.5 and 2 the loop stays stable; these runs start from the CTR = 1 operating point, which shows how long the integrator needs (several ms) to recover from a large disturbance. Runs at 18 V and 30 V input and with the 22/100 Ω settings give the same regulation (Vout 12.22–12.23 V at 1 A).

## 5.6 Cold start

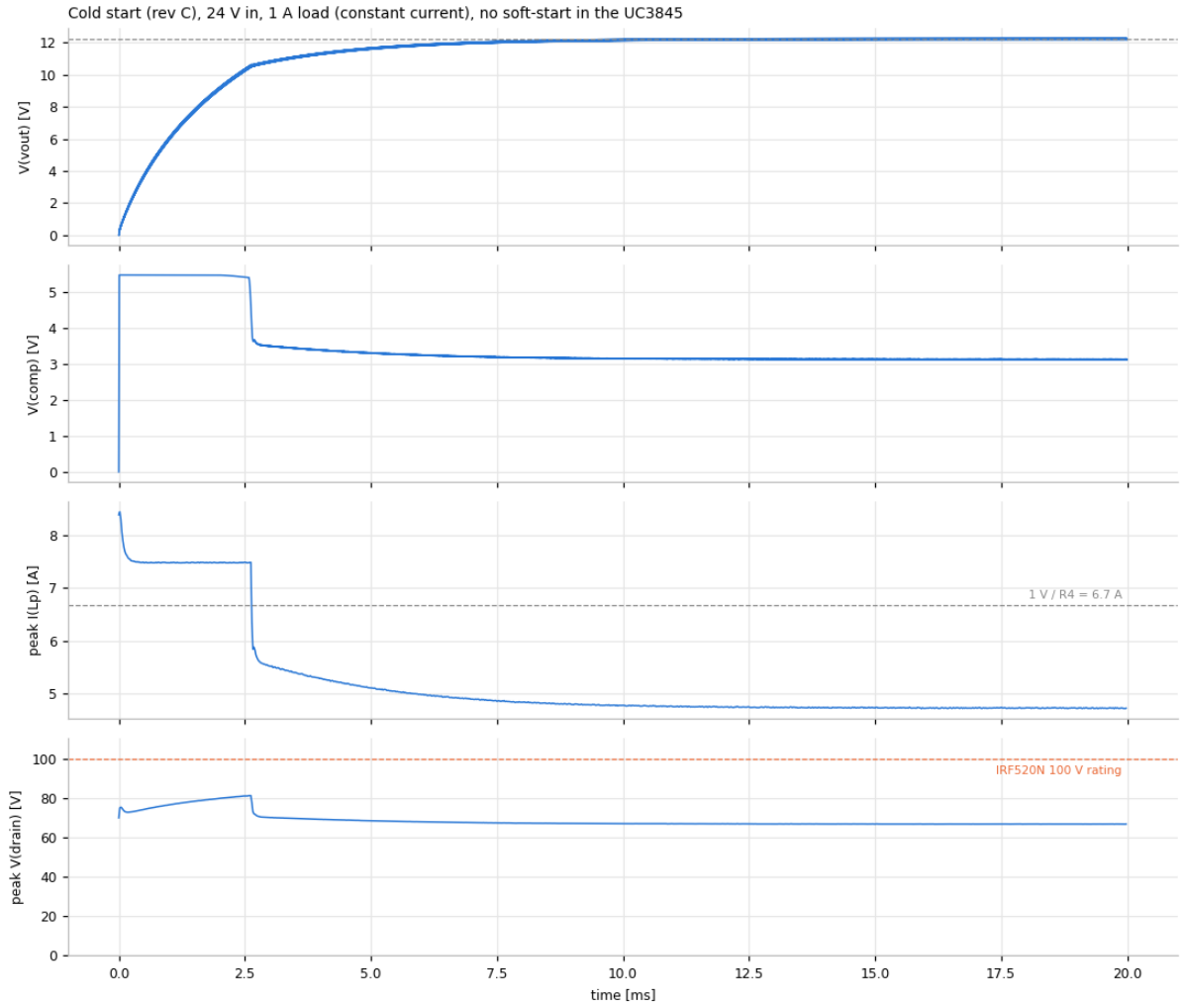



**Figure 7.** Cold start from 0 V with a 1 A constant-current load: output voltage, COMP, cycle peak current and cycle peak drain voltage.

Without a soft-start, COMP rises to its 5.5 V maximum and the converter runs at the current limit for the first 2.7 ms: 7.5 A peak (8.5 A in the first cycles, while the output is still near 0 V and the core resets slowly). The higher leakage energy raises the clamp voltage, and the drain peaks at 81 V, still 19 V below the IRF520N rating. The output reaches 12 V after 6.8 ms without overshoot.

## 5.7 Calculation versus simulation

**Table 10.** Comparison of hand calculation and simulation (24 V, 1 A, Rg 4.7 Ω).

| Quantity | Calculation | Simulation | Remark |
|---|---|---|---|
| Switching frequency | 153.6 kHz | 152.5 kHz | Oscillator window of the model |
| Duty cycle | 0.235 | 0.244 | Losses not in the lossless DCM formula |
| Primary peak current | 4.59 A | 4.73 A | +3 % |
| Secondary peak current | 6.9 A | 8.3 A (4.1 A per diode) | Leakage overshoot (Fig. 4) |
| Primary / secondary RMS | 1.28 / 2.15 A | 1.43 / 2.12 A | |
| Drain peak | 66 V | 66.4 V | Clamp at ≈ 42 V above VBUS |
| Clamp loss | 1.19 W | 0.91 W | Formula is conservative |
| R4 / D3 + D7 loss | 0.25 / 0.55 W | 0.29 / 0.35 W | |
| Output ripple | ≤ 0.21 V (ESR bound) | 75 mV | Ceramics take the HF current |
| Current limit at start-up | 6.7 + 0.96 = 7.6 A | 7.5 A | Delay-induced overshoot |
| Loop crossover | 5.2 kHz (τ ≈ 31 µs) | Step settles in ≈ 40 µs | Consistent |

## 5.8 Thermal estimate of the hot components

To estimate the temperature of the parts with the highest losses before a prototype exists, a steady-state thermal model of the board was built. It uses the real copper geometry, extracted from the KiCad file, and compact models of the hot components. It was first run on rev B, where it showed the clamp resistors at ≈ 92 °C; rev C was designed with it (three clamp resistors on local copper, a second rectifier diode). Rev D put Q1 on the aluminium case; the rev E layout (same parts, compact power cell) is analysed below and compared with rev C.

### *Model*

- **Board.** Both copper layers on a 0.2 mm grid (500 × 380 cells per layer), with the copper assigned to its net. Copper conducts in-plane only between cells of the same net, so the clearance gaps between nets are respected. FR-4 conducts in-plane (0.8 W/mK) and through the board (0.3 W/mK). 1 oz (35 µm) copper. The 229 vias and through-hole leads connect the two layers (plated barrel, or solder-filled lead).
- **Surfaces.** Top: natural convection for a horizontal plate ($Nu = 0.54\ Ra^{1/4}$, local temperature rise) plus radiation to the room (solder mask, ε = 0.9). Bottom (rev D): the 8 mm air gap to the case floor conducts heat (the warm board is above the cooler floor, so there is no convection) and the board radiates to the floor (bare aluminium, ε = 0.09, worst case). The case is one isothermal node that loses heat from its walls (vertical plates), its floor (downward and upward facing parts outside the board) and by radiation. Still air, 25 °C.
- **Components.** R9, R10, R20, R4 (2512) and R13 (1206): hot spot to solder joints 11 K/W and 20 K/W (Vishay). D3, D7 and D1 (SMC): junction to cathode lead 10 K/W, to anode lead 40 K/W. U2: junction to tab 15 K/W. Q1: $R_{th,jc}$ 3.1 K/W, Sil-Pad 400 6 K/W, 0.5 K/W spreading into the floor, the case node, and the 12 mm bent drain lead (≈ 75 K/W) into the board.
- **Losses.** From the closed-loop simulation at 24 V, 1 A (Table 8, unchanged in rev D): clamp 0.91 W (0.30 W per resistor), D3 and D7 0.17 W each, R4 0.29 W, Q1 0.73 W; plus D1 0.26 W, U2 0.18 W, R13 0.15 W and an assumed 0.10 W in T1. The Q1 conduction loss follows $R_{DS(on)}(T)$ (+0.75 %/K); the forward drop of D1, D3 and D7 decreases by 1.5 mV/K and their leakage doubles every 10 K. For the parallel diodes, the hotter one takes more current (≈ 8 mA per K of temperature difference). The model iterates until losses and temperatures are consistent; the energy balance closes exactly (2.94 W).

### *Results*

**Table 11.** Predicted component temperatures for rev E (hot spot or junction; solder pad in brackets), with rev C for comparison.

| Component | Loss | Rev C, 4.7 Ω | Rev E, 4.7 Ω | Rev E, 22 Ω | Rev E, 100 Ω | Rev E, worst case (note) | Limit |
|---|---|---|---|---|---|---|---|
| Clamp resistors | 0.30 W each | 68 °C | 70 °C (67) | 70 °C | 71 °C | 88 °C | 155 °C; pads ≤ 110 °C |
| Rectifier | 0.17 W each | 49 °C | 51 °C (50) | 51 °C | 52 °C | 68 °C | 150 °C |
| Q1 junction | 0.73 / 0.97 / 1.92 W | 43 °C | 41 °C | 45 °C | 58 °C | 83 °C | 175 °C |
| Heatsink (rev C: SK 104; rev D: case) | — | 40 °C | 34 °C | 35 °C | 40 °C | 57 °C | touch: ≤ 60 °C preferred |
| R4 (current sense) | 0.29 W | 54 °C | 53 °C (51) | 54 °C | 58 °C | 76 °C | 155 °C |
| D1 (reverse polarity) | 0.26 W | 63 °C | 68 °C | 68 °C | 69 °C | 84 °C | 150 °C |
| U2 (L78L12) | 0.18 W | 47 °C | 48 °C | 48 °C | 50 °C | 67 °C | 125 °C |
| R13 (preload) | 0.15 W | 50 °C | 52 °C | 52 °C | 53 °C | 69 °C | 155 °C |

Note: worst case = 40 °C ambient, Rg = 100 Ω, $R_{DS(on)}$ at its 0.20 Ω maximum, Sil-Pad 8.5 K/W (low mounting pressure), diode $R_{thJL}$ 15 K/W, 2512 $R_{hs\text{-}sp}$ 15 K/W and convection 20 % lower. The rev C worst case was 85 °C for the clamp resistors and 83 °C for Q1.

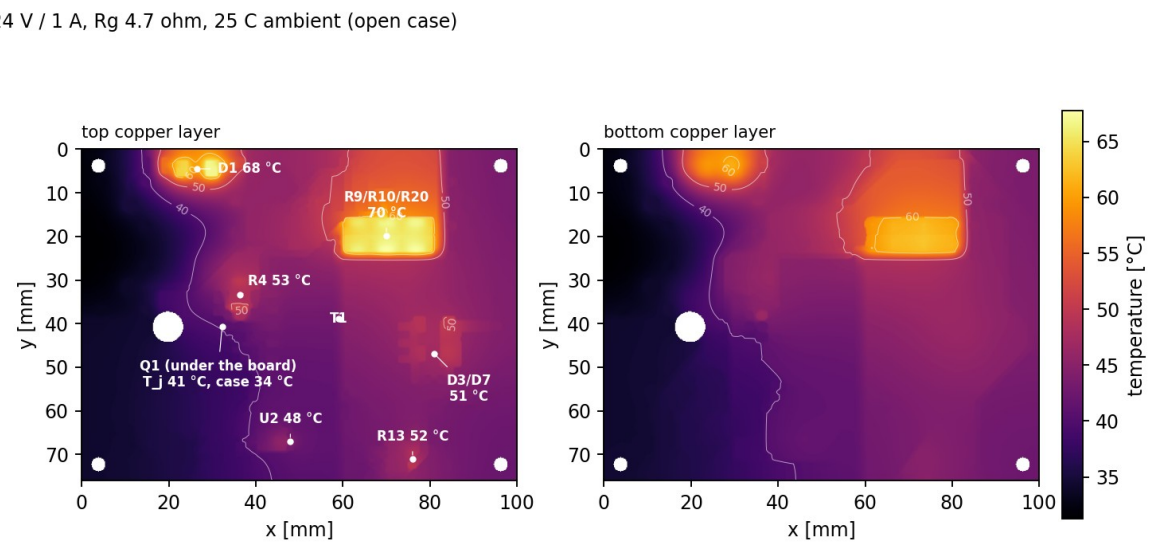



**Figure 8.** Predicted temperature of both copper layers of rev E at 24 V, 1 A, Rg = 4.7 Ω, 25 °C ambient, with the hot-spot or junction temperature of the main parts (Q1 sits under the board on the case).

**Interpretation.** Moving Q1 from the SK 104 to the case lowers its junction temperature from 43 °C to 41 °C at Rg = 4.7 Ω and from 63 °C to 58 °C at 100 Ω: the case (≈ 8 K/W) is a better heatsink than the SK 104 (≈ 17–19 K/W at these small temperature rises), even though the Sil-Pad (≈ 6 K/W) is a worse insulator than the silicone pad assumed for rev C. The rest of the board becomes 1–5 K warmer: its bottom side now faces the case floor, which is 9 K above ambient, through an 8 mm air gap (0.29 W flow from the board into the case). The clamp resistors (70 °C) and the reverse-polarity diode D1 (68 °C, little copper on the VF and VD nets) remain the hottest parts; D3 and D7 run at 51 °C with less than 1 K difference.

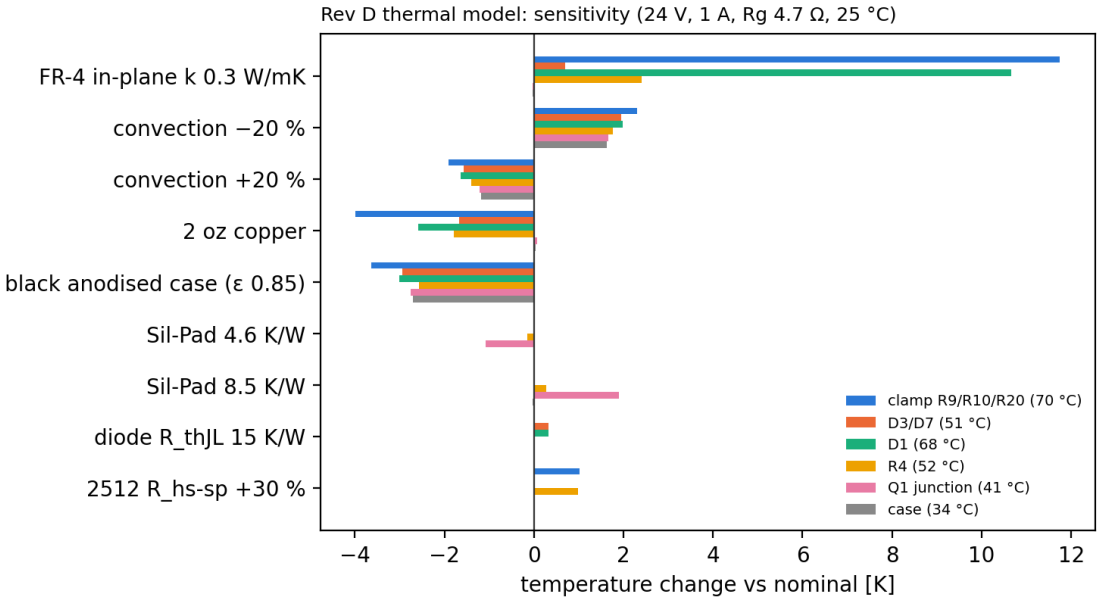



**Figure 9.** Sensitivity of the rev D/E hot-spot temperatures to the main model parameters (24 V, 1 A, Rg 4.7 Ω, 25 °C).

**Accuracy.** For Q1 the Sil-Pad dominates: 4.6–8.5 K/W moves the junction by about ±2 K at 4.7 Ω and ±5 K at 100 Ω; a black anodised case (ε = 0.85) halves the case thermal resistance (4.4 K/W) and lowers Q1 by 3 K and the clamp resistors by 4 K, because the board then also radiates into the floor (mask the MH5 contact spot when anodising). The convection coefficient (±20 % → ±2 K) matters for all parts. The largest uncertainty is still for the clamp resistors, whose heat partly crosses the gaps between nets through the FR-4 (in-plane conductivity of the laminate: up to +12 K, and +11 K for D1). The expected accuracy is about ±5 K for Q1, D3/D7 and R4 and about −5/+12 K for the clamp resistors. Not modelled: air flow through the open sides of the case, the transformer core loss (not known; 0.1 W assumed) and temperature differences within the case (it is treated as isothermal, ≈ 1 K error for 3 mm aluminium).

**Validation on the prototype.** (1) Measure the losses that are exactly measurable: $V(C15)^2/1.57\ k\Omega$ for the clamp, the RMS voltage across R4, and V·I of D3 and D7. (2) Calibrate the thermal resistances with known DC power and the converter off (current through the R9/R10/R20 group and through the Q1 body diode). (3) Measure the surface temperatures with 36 AWG type-K thermocouples bonded with thermal epoxy, or with an IR camera on a matt black dot, after 20–30 min, with the converter in its case on the plate in still air; compare D3 and D7 and measure the case next to Q1. (4) Use the Q1 body-diode voltage at 10 mA (≈ −2 mV/K after calibration) for its junction temperature. Then update the model parameters and re-run the worst case.

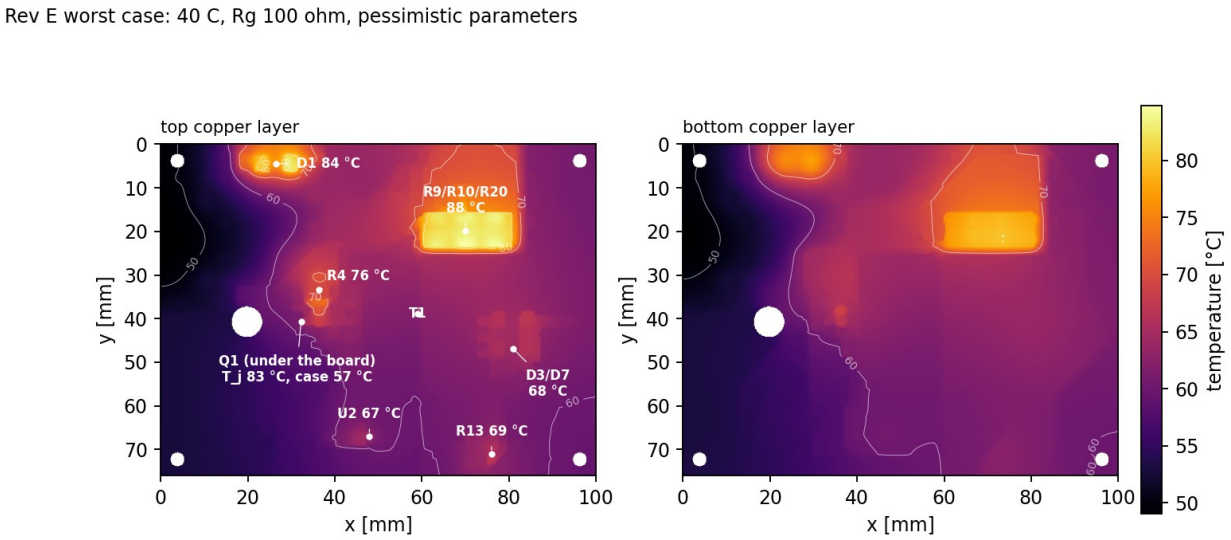



**Figure 10.** Rev E worst case: 40 °C ambient, Rg = 100 Ω and pessimistic thermal parameters.

# 6 PCB implementation and mechanics

## 6.1 Layout

The rev E layout is 100 × 76 mm, two layers, 1.6 mm FR-4, 1 oz copper. It is arranged in three bands. The top band holds the input sockets and the switchable EMI filter (J1 → F1 → D1 → L1 → C1–C3 → L2 → C4) and the clamp resistors. The middle band holds Q1 (under the board), T1 and the rectifier. The bottom band holds the UC3845 controller, the opto and the TL431 feedback. The input sockets J1/J2/J5 sit on the left edge and the output sockets J3/J4 on the right edge, matching the two walls of the case.

- The primary hot loop (C5/C6 → T1 → Q1 → R4) and the secondary loop (T1 → D3/D7 → C20/C21) are hand-routed and locked. The rest was autorouted with Freerouting.
- **Rev D changes.** Q1 is a through-hole part mounted from the bottom: its leads are bent up by 90° at 3.8 mm from the body and soldered on the top side, so that the body lies on the case floor (in rev E, Q1 sits directly below T1 pin 4, see the high-frequency loops below). The drain enters its pad straight from the hot-loop track. A 7 mm hole above the tab gives access to the M3 screw; it has a 1 mm copper-free ring because the screw and the screwdriver are at case potential. The SK 104 heatsink and its keep-outs are gone. MH5 is a plated M3 hole on the HS net for the metal stand-off to the case; MH1–MH4 are plain M3 holes (nylon stand-offs, no copper).
- **Right-angle sockets.** Cliff FCR7350 (24 A, 1 kV CAT III), contact on a 2.2 mm plated hole and four soldered fixing pins. The socket block overhangs the PCB edge by 2.5 mm so that its barrel passes through the case wall. J1/J2 keep the 19.05 mm (0.75 in) pitch of dual banana plugs; J3/J4 are 17.0 mm apart because D7 limits J4. J3 sits right of the isolation boundary (≥ 3.6 mm from the primary copper).
- Rev C copper for heat spreading is kept: CLAMP and VBUS areas with thermal vias under R9/R10/R20, and a 3 mm VOUT strip for D3/D7.
- A 3 mm copper-free band separates the PGND and SGND pours; only T1, U4, C23/C24 (YPS) and the chassis track cross it. The minimum primary–secondary copper distance, enforced by the design rules below, is 2.5 mm (at the clamp-resistor pads and around the YPS capacitor copper).
- No ground pour on the LISN side of the CM choke, so the pour cannot bypass the filter.
- Top and bottom pours stitched with vias on a 4 mm grid.
- Silkscreen: functional references for all jumpers and test points (DRAIN, GATE, CS, … ; CM1, RG, CASE, SNUB, …) and the jumper settings printed next to RG, CASE and YPS.
- Electrical checks: schematic and PCB netlists match exactly; DRC with the voltage-based rules below: 0 clearance errors and 0 unconnected items (remaining warnings are cosmetic silkscreen overlaps and the intended courtyard overlaps in the power cell (D2/C15/C5/C6/R4, SNUB/C16/R11), D3/D7 and the socket blocks over the corner screw heads).

### *Clearance and creepage (rev E)*

The board carries several potentials: the drain and the RCD clamp up to ≈ 66–73 V (100 V is used as the ceiling, the $V_{DS}$ rating of Q1), the input side at ≤ 30 V, the control circuits at ≤ 13 V, a floating secondary and the chassis/case. The spacing is therefore set per net from its worst-case voltage, not from one global clearance. On the flat board surface, without slots, the creepage path is the same straight line as the clearance, so one distance covers both. Solder mask is not counted as insulation.

**Table 12.** Voltage classes and design rules (net classes in emc_flyback.kicad_pro, domain rules in emc_flyback.kicad_dru).

| Class | Nets | Worst-case voltage | Clearance = creepage | Basis |
|---|---|---|---|---|
| P_HV | SW, SNA, CLAMP, AUX_A/B (unused T1 winding), HS, HSY (case) | 31–100 V (to PGND) | 0.6 mm | IPC-2221B A6 (bare terminations, 51–100 V): 0.5 mm |
| P_PWR, P_HI, CHASSIS | VIN, filter nets, VBUS, PSY1/2, PGND, CSRAW, CHASSIS | ≤ 32 V | 0.4 mm | IPC-2221B A6 (31–50 V): 0.4 mm |
| S_HI | SEC, VOUT, SGND | −20…+14 V (34 V apart) | 0.4 mm | IPC-2221B A6: 0.4 mm |
| P_SIG, S_SIG | gate drive, controller, feedback | ≤ 14 V | 0.2 mm | IPC-2221B A6 (≤ 15 V): 0.13 mm; fabrication |
| Primary ↔ secondary | every primary net to every secondary net | ≤ 100 V (floating) | 2.5 mm (3 mm band) | functional insulation (both sides SELV); ≥ IEC 60664-1 basic creepage 1.4 mm at 100 V, PD2, material group IIIb |
| Chassis ↔ secondary | chassis track to SGND/VOUT copper | ≤ 100 V (floating) | 1.5 mm | as above |

- **Enforced in KiCad.** The net classes and the domain rules are part of the KiCad project, so the KiCad DRC and the zone filler apply them; the pours pull back automatically (0.6 mm around the 100 V nets, 2.5 mm from primary copper, 1.5 mm from the chassis track). Exceptions: inside the three selector headers CASE, YPS and SGCH, whose 2.54 mm pins bridge domains on purpose (0.84 mm between pads, above the 0.5 mm IPC value at 100 V), and the L78L12 SOT-89 package (0.35 mm between the VBUS and PGND pads; the IPC value for ≤ 30 V is 0.25 mm).
- **Verified independently.** A script (check_spacing.py) assigns a voltage range to every net, measures the smallest copper-to-copper distance of all 1459 net pairs on both layers from the KiCad file (pads and vias treated as bare, tracks and pours as coated) and compares it with IPC-2221B and with the rules above: 0 pairs below IPC-2221B, 0 below the design rules. Smallest distances: 0.6 mm around the 100 V nets, 2.5 mm primary–secondary, 1.5 mm chassis–secondary.
- **Changes this required.** The chassis track was moved out along the right edge and away from the SGCH pin; the SGND link to SGCH was narrowed; the CLAMP via was moved from between the D2 pads (0.4 mm to the drain pad) to above D2; the VBUS bar was moved 0.25 mm away from the D2 cathode; the copper around the CASE header and the chassis track on the left were re-routed; the CLAMP heat-spreading copper was pulled back 0.7 mm; three SGND stitching vias beside the YPS capacitor copper were removed. Re-running the nominal thermal case on the changed copper gives +0.8 K on the clamp resistors (71 °C) and < 0.3 K elsewhere, so the thermal tables of Section 5.8 still apply.
- **Inside the parts.** Insulation across T1 (interwinding), U4 (LTV-817S, 5 kV), C23/C24 (2 kV, 1.8 mm between their pads) and the Q1 Sil-Pad (0.23 mm, 3.5 kV) is set by the components and their datasheets, not by the layout.

### *High-frequency loops (rev E)*

Rev E re-placed the power cell for the three loops that carry the fast edges, keeping all parts and values:

- **RCD clamp at the switch.** D2 and C15 sit directly between the T1 primary pins (1/2 = VBUS, 3/4 = drain) and the input capacitors C5/C6, with Q1 straight below T1 pin 4 and R4 under the Q1 source. At turn-off the leakage current commutates from Q1 into D2/C15 and returns through C5/C6 → R4 → Q1; this loop decides the voltage spike that the clamp does not catch ($V = L\,di/dt$, ≈ 13 V for 14 nH and 4.7 A in 5 ns).
- **RC snubber at the drain.** SNUB (on the drain track), C16 and R11 form a 7 mm column between the drain and the PGND end of R4, so the snubber current no longer crosses the board: in rev D it sat 25 mm away, at the end of a long drain track whose inductance was in series with the damping resistor.

- **Secondary rectifier.** D7 and D3 sit right next to T1 pins 7-9, C20/C21 next to the diode cathodes, and their SGND pads return on a 1.5 mm strip and the continuous bottom SGND plane straight back to T1 pins 10-12.

The loop inductances below are estimates from the track centre lines (Neumann double integral over the closed path, 0.35 mm equivalent conductor radius, no ground-plane image), so they are upper bounds and best used for comparison; the solid bottom planes under both loops lower the real values further.

**Table 13.** Estimated loop area and inductance, rev D → rev E.

| Loop | Rev D area | Rev D L | Rev E area | Rev E L |
|---|---|---|---|---|
| primary power loop (C5/C6, T1 1-4, Q1, R4) | 98 mm² | 25 nH | 81 mm² | **18 nH** |
| clamp commutation loop (Q1, R4, C5/C6, C15, D2) | 62 mm² | 26 nH | 50 mm² | **14 nH** |
| RCD clamp loop (D2, C15, T1 1-4) | 18 mm² | 7 nH | 16 mm² | **6 nH** |
| drain RC snubber loop (SNUB, C16, R11, R4, PGND) | 119 mm² | 32 nH | 46 mm² | **13 nH** |
| secondary loop via D7 (T1 7-9, D7, C20/C21, SGND) | 106 mm² | 27 nH | 49 mm² | **15 nH** |
| secondary loop via D3 | 172 mm² | 34 nH | 119 mm² | **25 nH** |

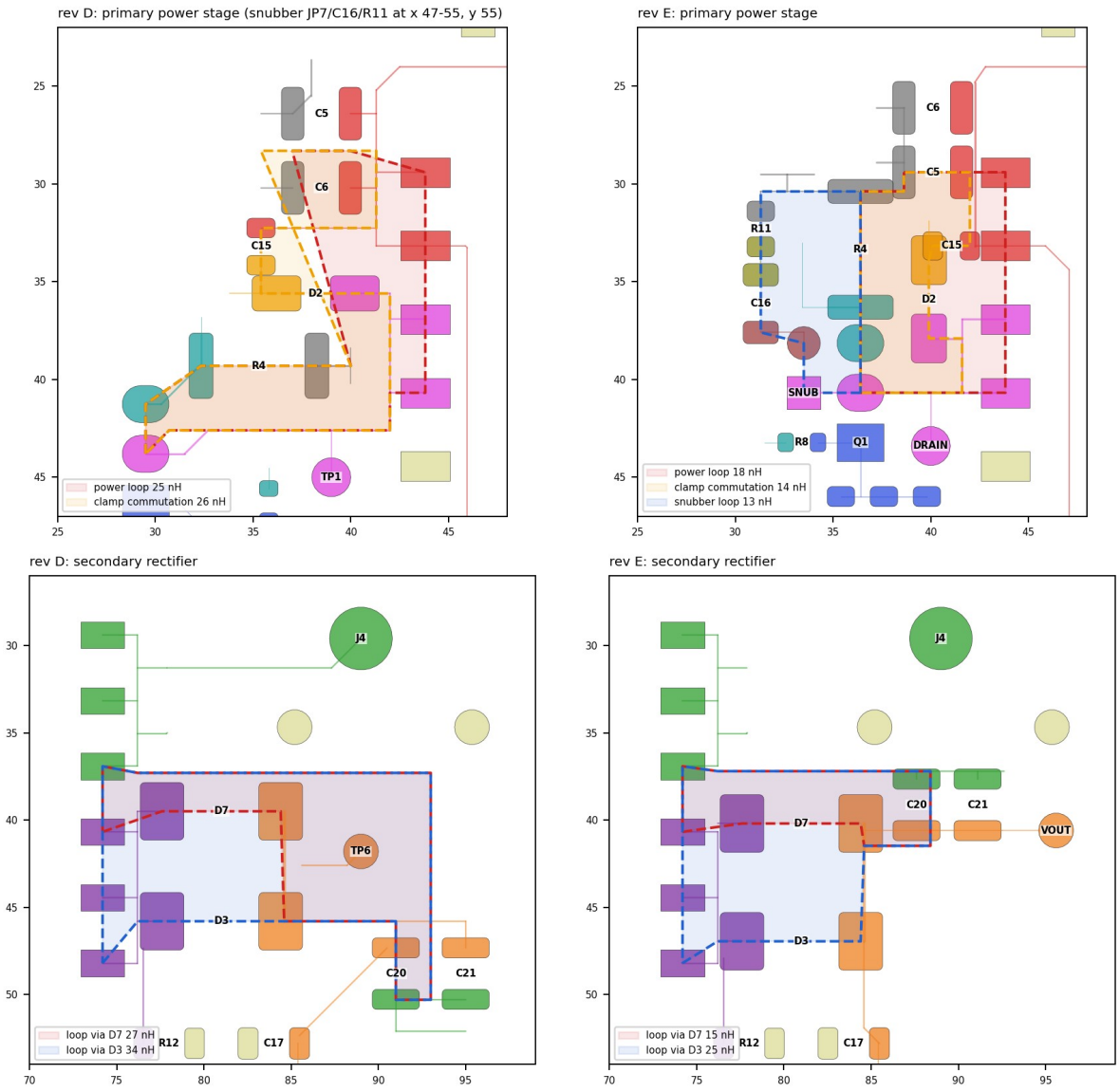



**Figure 11.** Power cell of rev D (left) and rev E (right), top-layer pads and tracks with the estimated loops: primary power loop, clamp commutation loop and RC snubber loop (top); secondary rectifier loops through D7 and D3 (bottom).

SW copper fell from 87 to 53 mm², which also lowers the drain-to-plane and drain-to-case capacitance that drives the common-mode current (Section 6.3). The price is a denser power cell: D2/C15/C6 and SNUB/C16/R11 overlap their courtyards by design, and these parts are hand-soldered or placed by JLC with the given CPL.

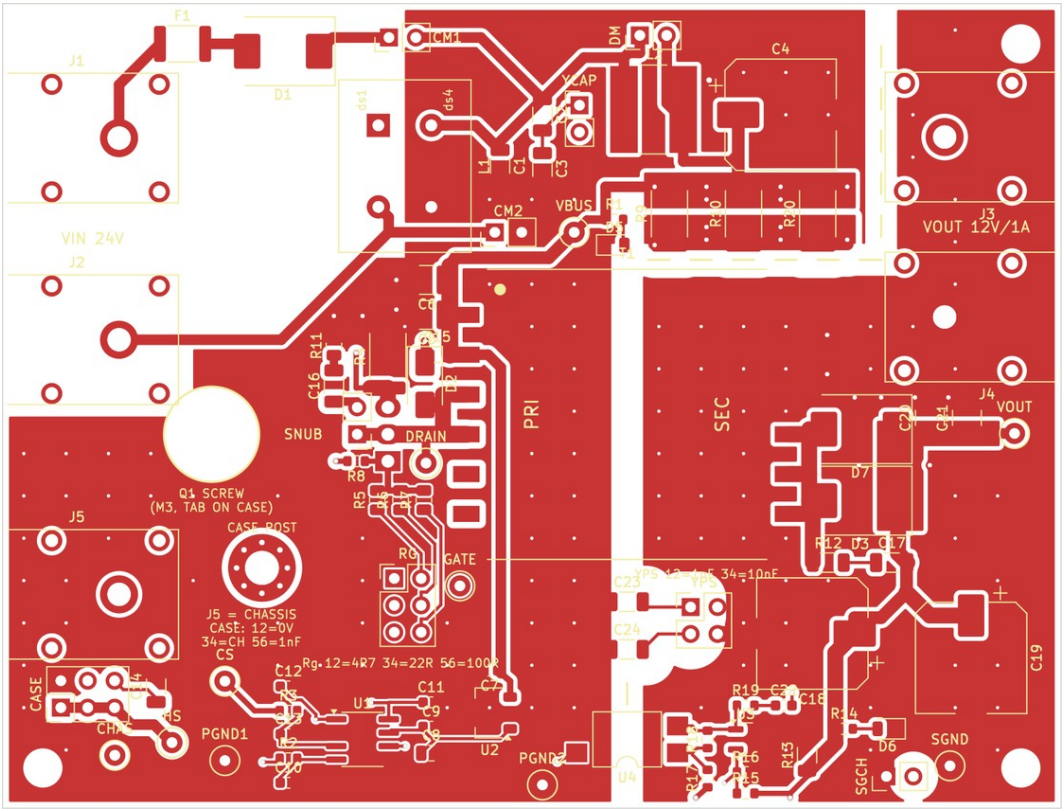



**Figure 12.** Rev E PCB, top copper and silkscreen (100 × 76 mm). The 7 mm ring left of the middle is the access hole for the Q1 screw; MH5 (CASE POST) is below it.

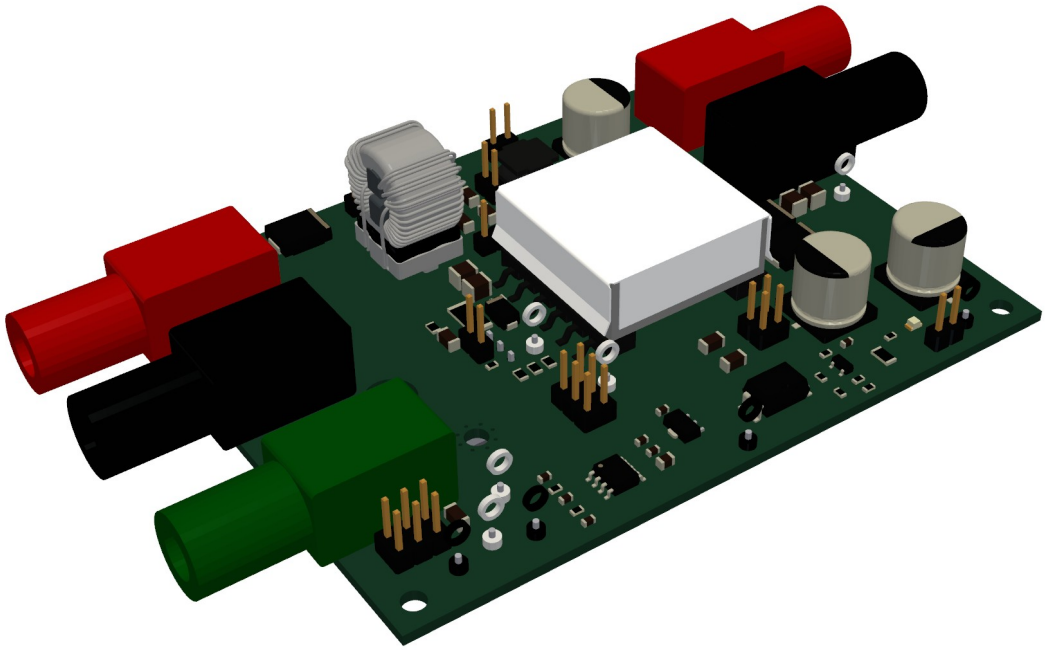


**Figure 13.** 3D view of the assembled rev E board (manufacturer STEP models for T1, L1 and L2; simplified sockets, test points and Q1 under the board).

## 6.2 Aluminium case (Q1 heatsink) and assembly

The case is a half-open U-channel bent from 3 mm aluminium (5754 or 5052, which bend well; 6061-T6 cracks): a 112 × 94 mm floor and two 33 mm high end walls (30 mm inside), open along the long sides and closed at the top

by a transparent 3 mm acrylic (PMMA) cover. It is the heatsink of Q1 and, electrically, the 'heatsink/housing' of the lecture example. The workshop drawing and the assembly are shown below; the STEP model is in the case folder.

**Table 14.** Mechanical stack-up (heights above the top of the case floor).

| Item | Height / value | Comment |
|---|---|---|
| Sil-Pad 400 under the tab | 0 – 0.23 mm | Bergquist SP400-0.009-00-54 (TO-220 pre-cut), 3.5 kV |
| Q1 tab / top of the body | 0.23 – 1.53 mm / 4.73 mm | TO-220 lying flat, tab down, marking towards the PCB |
| Q1 screw head (M3 pan + shoulder washer) | ≈ 1.5 – 4.3 mm | ≥ 3.7 mm below the PCB; fitted through the 7 mm hole |
| PCB bottom / top | 8.0 / 9.6 mm | M3 × 8 hex stand-offs (MH1–MH4 nylon, MH5 metal) |
| Socket axis | 15.1 mm | 5.5 mm above the PCB; wall slots 12 mm wide, open at the top |
| Q1 lead length used | 11.7 mm | 3.8 mm horizontal + vertical through the PCB; IRF520N leads ≥ 12.7 mm |
| Wall top = underside of the cover | 30 mm | L1 (tallest part) ends at 27.6 mm: 2.4 mm clearance |
| Cover (clear PMMA) | 30 – 33 mm | On the wall tops and four M3 × 30 hex pillars in the corners; overall height 36 mm with the floor |

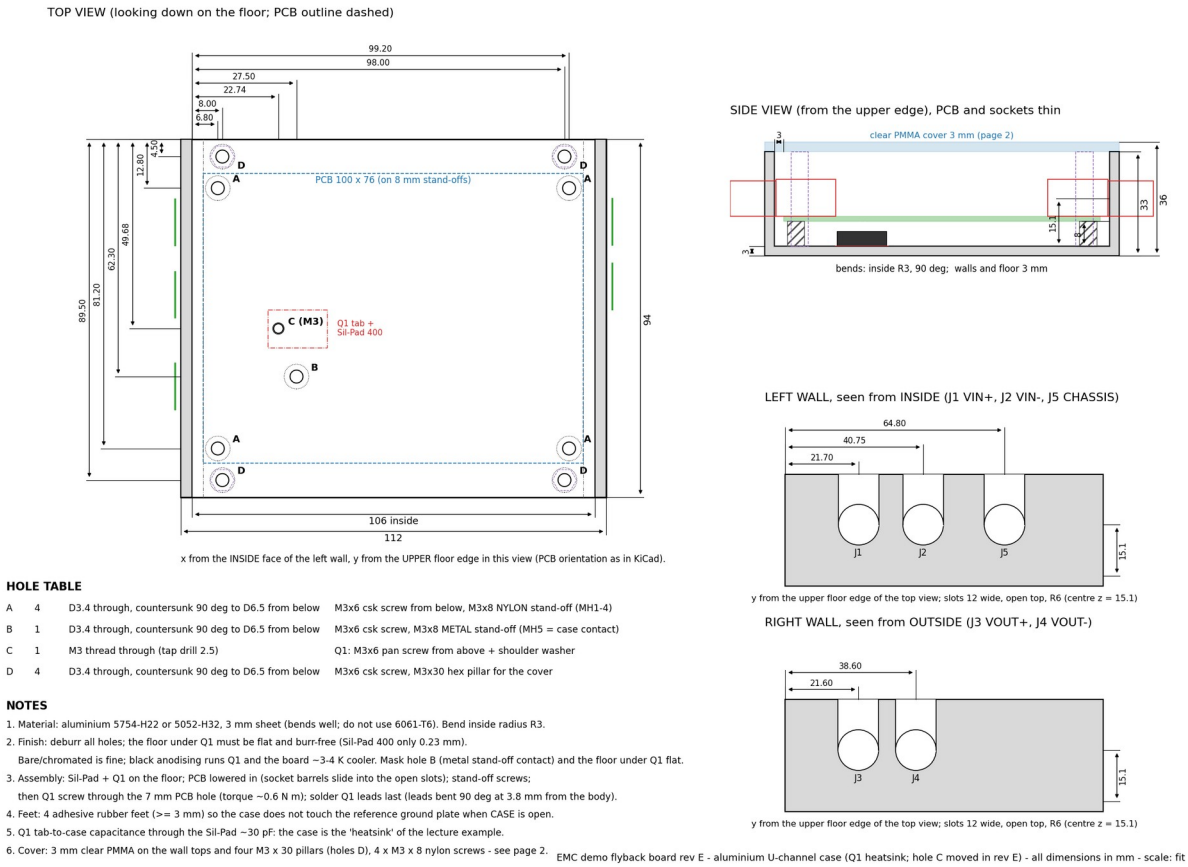



**Figure 14.** Workshop drawing of the aluminium U-channel case (hole table and notes included; page 1 of the PDF in the case folder).

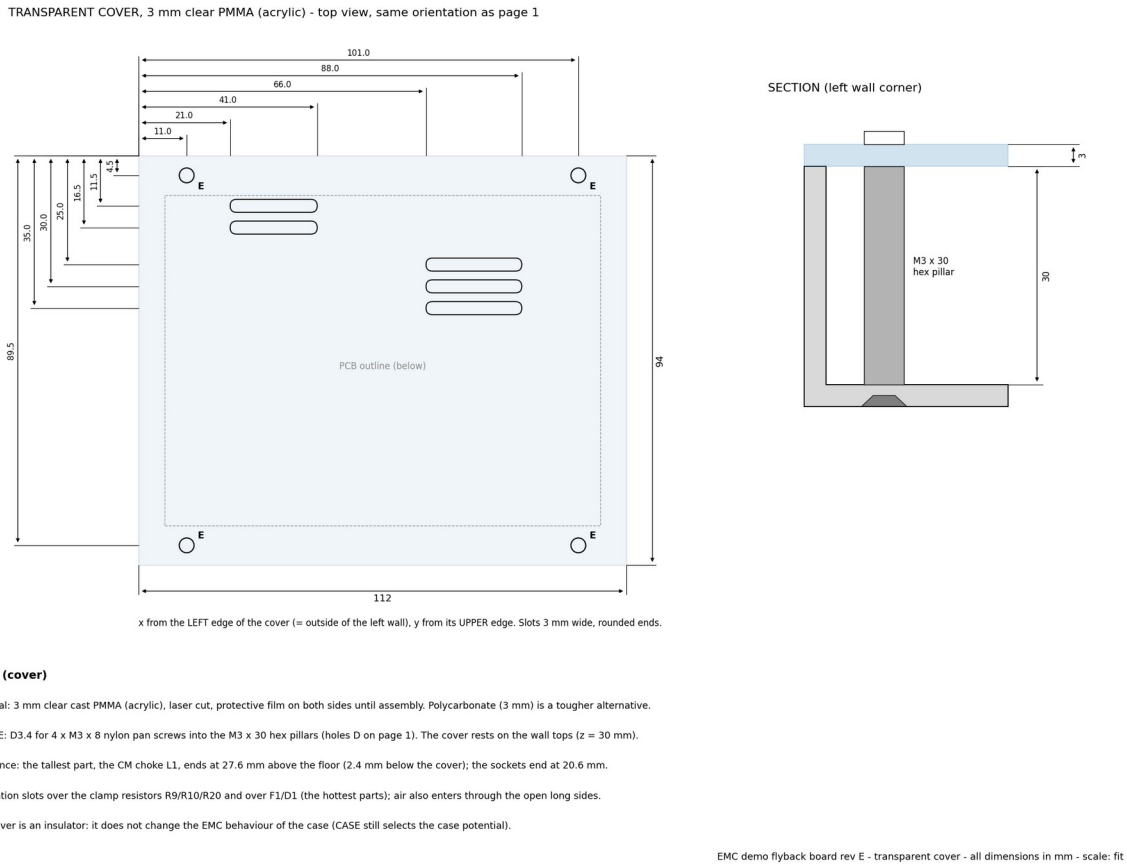



**Figure 15.** Transparent cover: 3 mm clear PMMA, four holes for the M3 pillars and five ventilation slots over the clamp resistors and F1/D1 (page 2 of the PDF).

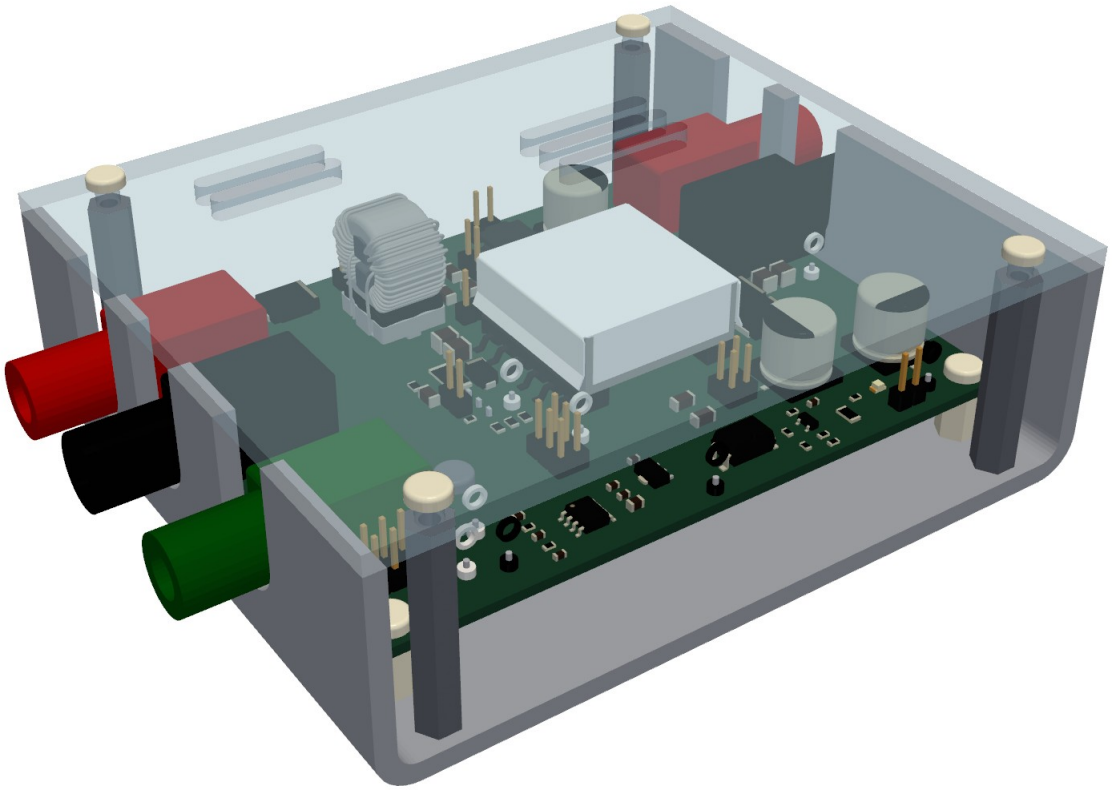


**Figure 16.** The rev E board in its case with the transparent cover. The case, stand-offs, pillars, screws and feet are the 3D model of the board-only footprint MECH1 and the cover that of MECH2 (drawn with 30 % opacity), so they also appear in the KiCad 3D viewer. The socket barrels pass through the open slots in the end walls; Q1 is under the board.

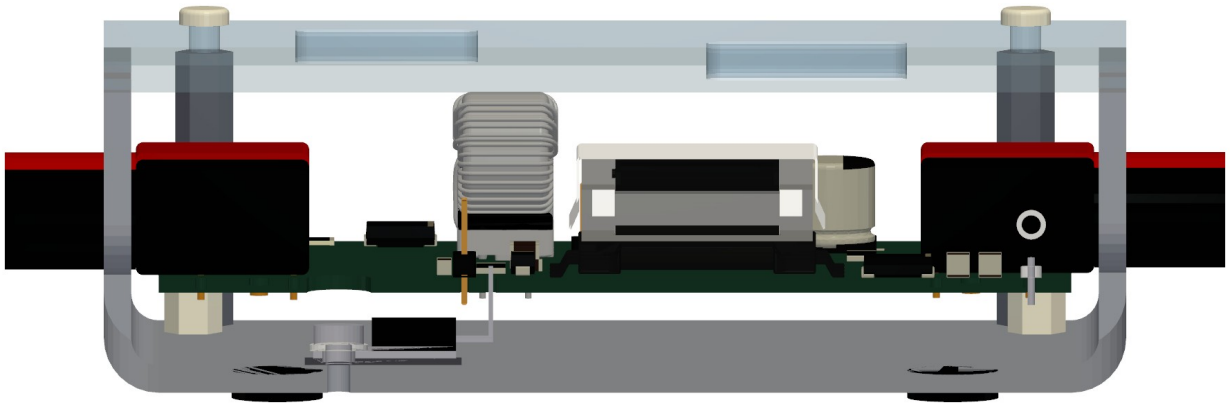


**Figure 17.** Section through the Q1 screw: Q1 lies on the case floor (Sil-Pad, shoulder washer, M3 screw), its leads are bent up into the PCB; the PCB stands on 8 mm stand-offs; the sockets J2 (left) and J4 (right) pass through the walls; the cover sits on the walls and the corner pillars.

**Assembly order.** (1) Bend the Q1 leads 90° at 3.8 mm from the body (pliers or a bending jig, never at the body). (2) Fit the stand-offs to the case floor with countersunk M3 screws from below (MH5 metal, the others nylon) and stick on the four rubber feet. (3) Put the Sil-Pad and Q1 on the floor and fit the M3 × 6 screw with the shoulder washer loosely. (4) Lower the PCB, with all other parts already soldered, so that the socket barrels slide into the wall slots and the Q1 leads enter their holes; screw the PCB to the stand-offs. (5) Tighten the Q1 screw through the 7 mm hole (≈ 0.6 N m). (6) Check > 10 MΩ between the tab (DRAIN test point) and the case, ideally with a 500 V insulation tester. (7) Solder the Q1 leads from the top.

**Transparent cover.** The cover lets the students see the board while keeping fingers off the 66 V drain node and the hot clamp resistors. It is laser cut from 3 mm clear cast acrylic (polycarbonate is a tougher alternative), rests on the wall tops and is held by four M3 × 8 nylon screws in M3 × 30 aluminium hex pillars that stand in the 9 mm strips of floor in front of and behind the PCB (holes D). It is an insulator, so it does not change the EMC behaviour of the case. Fitting it is the last assembly step.

**Thermal effect of the cover.** The cover blocks most of the convection from the top of the board. The thermal model was re-run with the board top exchanging heat only with the cover across the 20 mm air layer (enclosed-layer convection plus radiation) and the cover losing heat to the room; the five ventilation slots, the open long sides and the conduction into the walls were ignored, so these are upper bounds:

**Table 15.** Predicted temperatures with the closed cover (vents ignored) compared with the open case.

| Part | Open, 4.7 Ω | Cover, 4.7 Ω | Cover, 100 Ω | Cover, worst case | Limit |
|---|---|---|---|---|---|
| Clamp resistors, hot spot (pad) | 70 °C | 87 °C (84) | 89 °C | 104 °C (101) | 155 °C; pads ≤ 110 °C |
| D1 | 68 °C | 83 °C | 85 °C | 98 °C | 150 °C |
| D3/D7 | 51 °C | 66 °C | 69 °C | 83 °C | 150 °C |
| R4 | 53 °C | 67 °C | 73 °C | 90 °C | 155 °C |
| U2 | 48 °C | 62 °C | 65 °C | 81 °C | 125 °C |
| Q1 junction / case | 41 / 34 °C | 45 / 36 °C | 62 / 42 °C | 87 / 60 °C | 175 °C |
| Cover | — | 36 °C | 37 °C | 53 °C | PMMA ≤ 70 °C |

The parts on the board become 13–17 K warmer; Q1 hardly changes because its heat goes into the case. Everything stays within its rating even in the worst case, but the clamp-resistor pads approach the 110 °C solder-joint guideline at 40 °C ambient and C4 (electrolytic, next to the clamp resistors) will age faster. The ventilation slots above R9/R10/R20 and F1/D1 and the open long sides will recover part of the difference; for long demonstrations at 100 Ω or in a warm room, run without the cover or check the clamp resistors with an IR thermometer through the slots.

Rev E with closed cover (vents ignored), Rg 4.7 ohm, 25 C

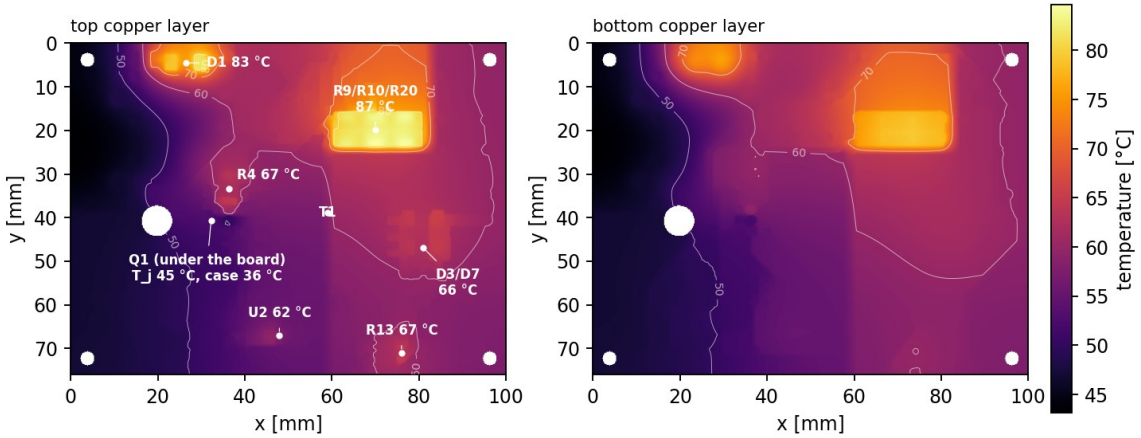



**Figure 18.** Predicted board temperature with the closed cover (vents and side openings ignored), 24 V, 1 A, Rg = 4.7 Ω, 25 °C.

**Socket forces.** The FCR7350 flange is only 0.25 mm wider than the wall slot on each side, so the wall does not take the plug forces: they go into the PCB through the contact and the four soldered fixing pins. Hold the socket when pulling a stiff plug.

## 6.3 The case as an EMC knob

The drain is connected to the TO-220 tab, so the full switch-node voltage drives a common-mode current into the case through the Sil-Pad:

$$C_{tab} = \frac{\varepsilon_0 \varepsilon_r A}{d} = \frac{8.85\,pF/m \cdot 5.5 \cdot 140\,mm^2}{0.23\,mm} \approx 30\,pF\,,\; I_{CM} = C_{tab}\frac{dv}{dt} \approx 150\,mA\;at\;5\,V/ns \qquad (19)$$

Where this current goes depends on the CASE jumper. With CASE = CHASSIS it flows from the case through J5 to the reference plate and back through the two LISNs, which see it as common-mode noise: $V_{LISN} = I_{CM} \cdot 25\ \Omega$ (the two 50 Ω LISN impedances in parallel). With CASE open the case floats and the path is closed only through the case-to-plate capacitance (≈ 28 pF on 3 mm feet), which roughly halves the current. With CASE = PGND (or 1 nF, which is a short above ≈ 100 kHz compared with $C_{tab}$) the current returns to the source of the noise, the primary ground, without passing the LISNs. The figure below estimates the LISN voltage from the simulated $V_{DS}$ harmonics for the first two cases.

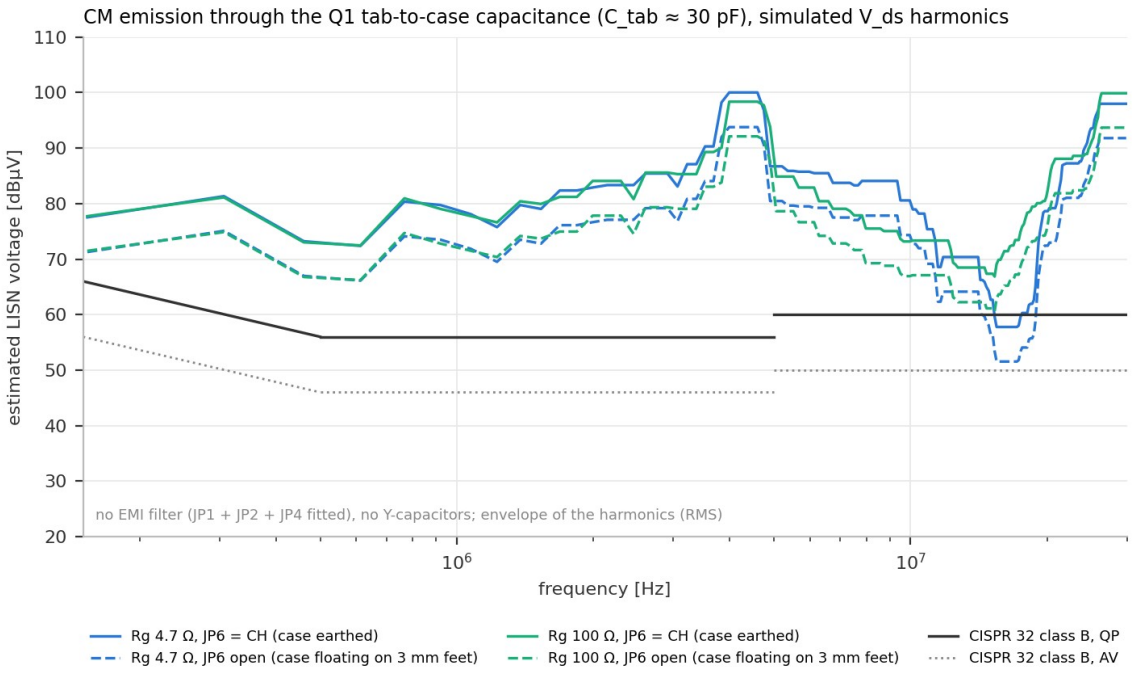



**Figure 19.** Estimated common-mode LISN voltage caused by the Q1 tab-to-case capacitance alone (no EMI filter, no Y-capacitors, no transformer capacitance). CISPR 32 class B limits for orientation.

Without any filter this one path is 20–40 dB above the class B quasi-peak limit, which shows why the heatsink connection is such an effective knob: CASE = PGND removes it, while the CM choke L1 and the Y-capacitors (CM1/CM2/YCAP) must otherwise provide the attenuation. The estimate ignores the transformer interwinding capacitance, the LISN and cable resonances and the board's own capacitance to the case floor (a few pF for each ground pour at 8 mm), so it shows trends and orders of magnitude, not exact levels. A thicker insulator (e.g. 0.5 mm, ≈ 14 pF) or a shielded insulator with its shield on PGND are further variants to try.

### *Effect of the case capacitance on the losses*

With CASE = PGND (or 1 nF, or CHASSIS at high frequency) the ≈ 30 pF tab capacitance appears in parallel with Q1, from drain to PGND. This adds to the ≈ 75 pF already on the drain node (Q1 $C_{oss}$, D2, T1), so it can change the switching losses. The closed-loop simulation (ngspice, the netlist of the LTspice model) was repeated with 30 pF and, as a margin for Sil-Pad tolerance, 60 pF from drain to PGND:

**Table 16.** Simulated effect of the drain-to-PGND case capacitance at 24 V, 1 A (closed loop, ngspice).

| Case | Q1 loss | RCD clamp loss | $V_{clamp}$ / $V_{DS,max}$ | Input power | Efficiency |
|---|---|---|---|---|---|
| Rg 4.7 Ω, no extra C | 0.734 W | 0.907 W | 37.6 V / 66.4 V | 14.660 W | 83.47 % |
| Rg 4.7 Ω, + 30 pF | 0.702 W (−32 mW) | 0.916 W (+10 mW) | +0.2 V / +0.2 V | −23 mW | 83.60 % |
| Rg 4.7 Ω, + 60 pF | 0.693 W (−41 mW) | 0.926 W (+19 mW) | +0.4 V / +0.4 V | −22 mW | 83.59 % |
| Rg 100 Ω, + 30 pF | 1.901 W (−16 mW) | 0.878 W (+4 mW) | +0.1 V / 0 V | −11 mW | 77.27 → 77.32 % |

- **Turn-off.** The extra capacitance takes up to ≈ 0.4 A of the 4.7 A while the drain voltage rises, like a lossless snubber: the edge is slightly slower (9.5 → 8.7 V/ns) and the voltage–current overlap in Q1 is smaller. The turn-off energy falls from 2.85 µJ to 2.63 µJ (30 pF) and 2.59 µJ (60 pF), ≈ −35 mW at 153 kHz. At Rg = 100 Ω the gate already sets a slow edge, so the gain is smaller.
- **Turn-on.** The stored energy ½·C·V² is dissipated in Q1. In DCM the drain is only at the residual ringing voltage when Q1 turns on (≈ 5–43 V, depending on where in the ringing the next cycle starts), not at 66 V: 3 → 20 nJ (30 pF, turn-on at 7 V) and 94 nJ (60 pF, turn-on at 25 V), i.e. 0.5–14 mW; ≈ 4 mW for 30 pF at the worst ringing peak (43 V).

- **RCD clamp.** Slightly more energy ends up in the clamp: $V_{clamp}$ rises by 0.2 V and its loss by ≈ 1 % (+10 mW); the drain peak stays at ≈ 66.6 V, well below the 100 V rating.
- **Ringing and current sense.** The drain-node capacitance grows from ≈ 75 to ≈ 105 pF: the ringing after the clamp interval moves from 28.8 to 24.3 MHz (21.5 MHz with 60 pF) and the DCM ringing from 4.2 to 3.9 MHz, which shifts the 20–30 MHz peak in the spectrum slightly down. The ≈ 0.15 A, ≈ 10 ns discharge spike at turn-on (≈ 20 mV on R4) is removed by the R3/C13 filter in front of the 1 V current-sense threshold.

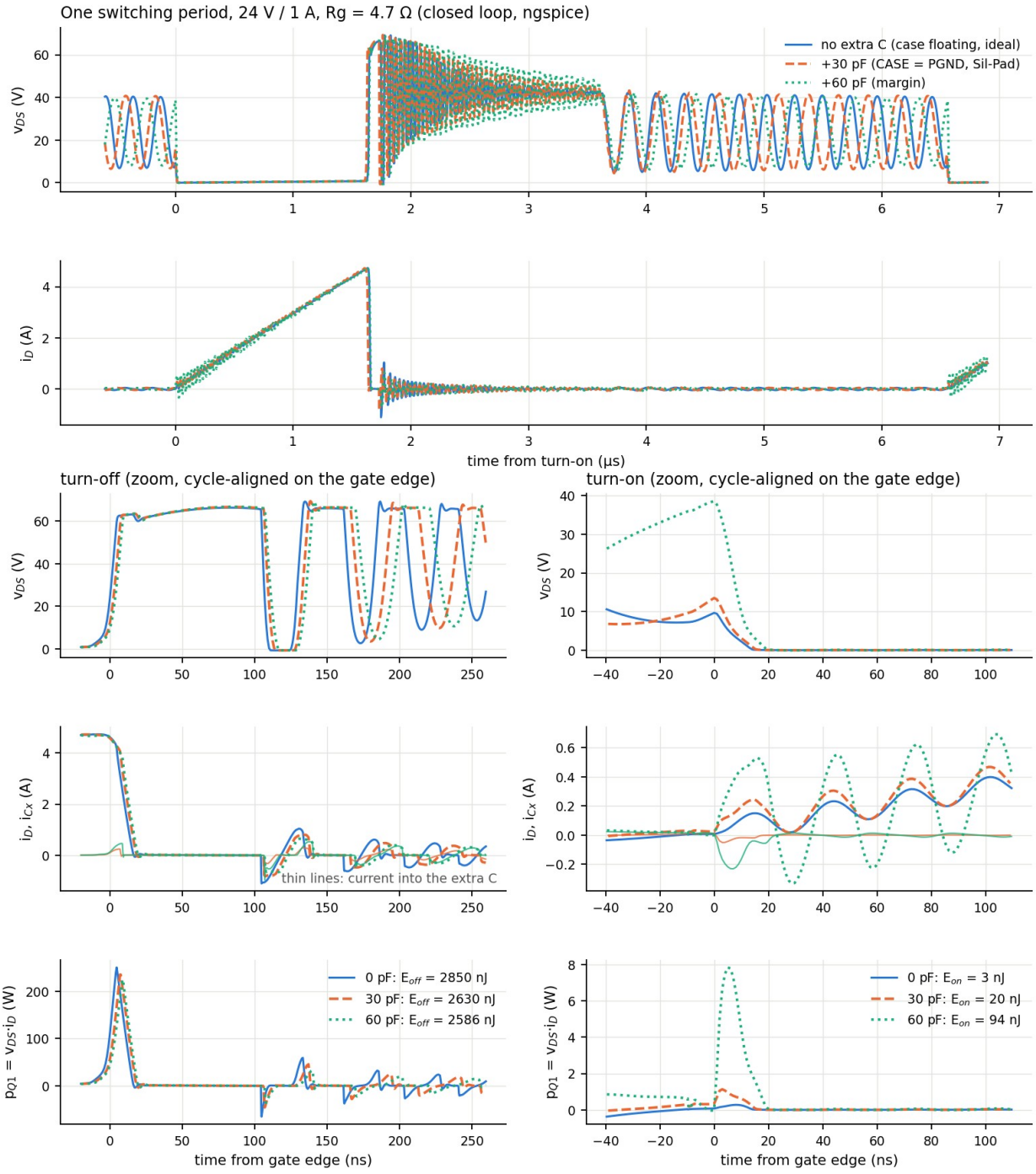



**Figure 20.** Q1 drain voltage, drain current and instantaneous power with 0, 30 and 60 pF extra drain-to-PGND capacitance (24 V, 1 A, Rg = 4.7 Ω, closed loop, ngspice; turn-off and turn-on aligned on the gate edge, energies averaged over 10 cycles). Thin lines: current into the extra capacitance.

The net effect is a slight improvement (−11 to −23 mW, ≤ 0.15 % efficiency), so the choice of the CASE setting is an EMC choice, not an efficiency one. With CASE = CHASSIS the same charge flows every cycle, but through the plane and the LISN 50 Ω, which is why it is so visible in the measurement.

# 7 Key components

**Table 17.** Key components (full list in Appendix B and Full_BOM.csv).

| Ref | Part | Value / rating | Why this part |
|---|---|---|---|
| U1 | UC3845BD1013TR | Current-mode PWM, 50 % max duty | Classic teaching controller, peak-current mode, 1 A driver |
| T1 | Würth 750311591 | 8 µH, 1.5:1:0.67, $I_{sat}$ 20 A, 1500 $V_{AC}$ | Off-the-shelf flyback transformer (EFD25 SMD) |
| Q1 | IRF520NPBF | 100 V, 9.7 A, 0.2 Ω, TO-220 | TO-220 under the board, tab on the case (Williams heatsink experiment) |
| Cover | 3 mm clear cast PMMA, laser cut | 112 × 94 mm, 4 holes, 5 vent slots | Touch protection; students can still see the board |
| Case | Bent 3 mm aluminium U-channel | 112 × 94 × 33 mm, ≈ 8 K/W (bare) | Q1 heatsink and the lecture's 'housing' |
| Q1 insulator | Bergquist Sil-Pad 400 (SP400-0.009-00-54) | 0.23 mm, 3.5 kV, 4.6–8.5 K/W | Defined $C_{tab}$ ≈ 30 pF; nylon shoulder washer for the M3 screw |
| J1–J5 | Cliff FCR7350R/B/G | 4 mm right-angle PCB socket, 24 A | Barrels through the case walls |
| D3, D7, D1 | SS510C | 100 V, 5 A Schottky (SMC) | No reverse recovery; low $V_F$; D3 ‖ D7 share the output current |
| R9, R10, R20 | 25121WJ0472T4E (UNI-ROYAL) | 4.7 kΩ 1 W 2512 | Clamp resistor split over three parts (LCSC number still to be looked up) |
| D2 | ES2D | 200 V, 2 A, 20 ns | Fast clamp diode |
| L1 | Würth 744822222 | 2.2 mH, 2 A CM choke | CM filter knob |
| L2 | Sunlord SWPA8040S100MT | 10 µH, 3.3 A | DM filter knob (check JLC stock) |
| U4 | LTV-817S | Opto, CTR 50–600 % | Standard isolated feedback |
| U3 | TL431AIDBZR | 2.495 V shunt reference | Error amplifier on the secondary |
| U2 | L78L12ACUTR | 12 V, 100 mA LDO (SOT-89) | Controller supply from VBUS |
| C18, C19 | RVT1E471M1010 | 470 µF 25 V electrolytic | Bulk output capacitance |
| C20, C21 | CL32A226KAJNNNE | 22 µF 25 V X5R 1210 | HF output current |
| R4 | HoLR2512-150mR-1% | 0.15 Ω 1 W | Current sense |

# 8 Bring-up and test plan

## 8.1 Bring-up

1. Fit the hand parts and assemble the board in its case as described in Section 6.2. Before soldering the Q1 leads, check > 10 MΩ between the tab and the case, and with CASE open between the case and every PCB net.
2. Settings: RG = 1-2 (4.7 Ω), CASE open, SNUB open, no load. Bench supply 24 V with a 0.3 A current limit. Check VCC (12 V), VREF (5 V), the RT/CT sawtooth (307 kHz), the gate (153 kHz) and VOUT (12.2 V).
3. Load 12 Ω / 15 W, raise the current limit to 1 A. Check $V_{DS}$ < 80 V, the clamp-resistor temperature (R9/R10/R20 ≈ 0.3 W each), that D3 and D7 run at about the same temperature (current sharing), and the case temperature next to Q1.
4. Load step 0.5 A ↔ 1 A (electronic load): expect ≈ ±25 mV deviation and no ringing on VOUT and COMP.
5. Cold start at 1 A: the CS (current-sense) test-point peak should stay ≤ 1.3 V (≈ 8.5 A).

## 8.2 EMC measurements: set-up

Conducted emissions are measured with two single-line 5 µH / 50 Ω LISNs, one in each supply lead, and an oscilloscope; the load is a 12 V LED strip or module of 10–12 W. Figure 21 shows the connections.

- **Plane and placement.** A copper or aluminium reference plane (≈ 1.0 × 0.6 m) on the bench, bonded to PE at one point: a short braid (≤ 30 cm) to the green/yellow ⏚ terminal of the bench PSU, which is earthed through its mains cord (not to the − output). The scope adds its own PE path through the coax shields; no further bonds (radiators, pipes, other sockets). Check < 1 Ω between the plane and a scope BNC shell. Both LISN cases are bolted to the plane. The converter (in its case, cover on) stands on a 50 mm insulating support; the leads LISN → J1/J2 are ≤ 0.5 m long, 50 mm above the plane and laid the same way every run. J5 (CHASSIS) is strapped to the plane with ≤ 10 cm of braid.

### *Earthing the reference plane (finding PE in a classroom)*

The plane needs exactly one low-impedance bond to protective earth (PE). In a lecture room there is usually no earth terminal on the wall, but the instruments brought in are earthed through their mains cords, so the bond is made to one of them:

**Table 18.** Ways to bond the reference plane to PE.

| Option | How | Notes |
|---|---|---|
| 1. PSU earth terminal (preferred) | Short braid or 4 mm lead (≤ 30 cm) from the plane to the green/yellow ⏚ binding post on the front of the bench PSU | The ⏚ post is wired to PE through the mains cord. Do not use the − output or a black "COM" terminal: the PSU outputs must stay floating. |
| 2. Oscilloscope ground | Functional-earth stud on the rear panel, or a BNC shell | The plane is already earthed through the coax shields (scope BNC → coax → LISN RF ground → LISN case → plane); the explicit bond makes this path defined and keeps it when the scope is unplugged. |
| 3. ESD earth plug | Earth-only plug for a Schuko (type F) socket with a 4 mm socket or 10 mm stud; only its earth clips are metal | Check the socket first with a plug-in socket tester (three-LED type) if the room wiring is doubtful. |

- **Check.** With a multimeter on the ohms range, measure between the plane and a scope BNC shell: it should read below about 1 Ω. Measure only between your own equipment, never into a mains socket.
- **Do not** use radiators, water pipes or cable conduit as earth, push wires into a socket or open a wall plate, or bond the plane at several points (PSU and scope and wall): one bond near the LISNs is enough, since the scope path through the coax is always present.
- **Supply and load.** Bench PSU 24 V with a 1.0 A current limit (≈ 0.6 A at full load), outputs floating (an earthed − terminal would bypass LISN − at RF), connected to the LISN SUPPLY ports. The LED load sits off the plane on J3/J4; a DMM in series is optional.
- **Scope.** The RF ports of LISN + and LISN − go through identical RG58 cables and 20 dB attenuators (or transient limiters, which protect the 50 Ω inputs at switch-on) to CH1 and CH2, set to 50 Ω input (with 1 MΩ inputs use 50 Ω feed-through terminators).
- **CM and DM.** Each LISN gives the noise of its lead to the plane: CH1 = $V_{CM}$ + $V_{DM}$, CH2 = $V_{CM}$ − $V_{DM}$. Scope math M1 = (CH1 + CH2)/2 gives the CM part and M2 = (CH1 − CH2)/2 the DM part; the FFT of M1 and M2 from 150 kHz to 30 MHz (100 MHz for the drain ringing) uses a Hann or flat-top window, ≥ 1 ms record (≤ 1 kHz bins), ≥ 250 MS/s and max-hold or 16× averaging, with deskewed channels at the same V/div. Level: dBµV = dBm + 107 (50 Ω) plus the 20 dB attenuator and the LISN factor. An FFT is a peak-type reading, not a CISPR quasi-peak or average receiver, so the settings are compared with each other (Δ dB).

**Table 19.** Baseline (reference) jumper settings for the EMC measurements.

| Jumper | Setting | Meaning |
|---|---|---|
| CM1, CM2 | open | CM choke L1 active |
| DM | open | DM inductor L2 active |
| YCAP | fitted | input Y-caps C2/C3 to chassis |
| RG | 1-2 | 4.7 Ω gate resistor (3-4 = 22 Ω, 5-6 = 100 Ω; fit exactly one) |
| CASE | 1-2 | case (Q1 heatsink) to PGND (3-4 CHASSIS, 5-6 1 nF, open = floating) |
| SNUB | open | no drain RC snubber |
| SGCH | open | output floating |
| YPS | open | no primary–secondary Y-cap (1-2 = 1 nF, 3-4 = 10 nF) |
| J5 | strapped | chassis socket to the plane |

## 8.3 EMC measurements: procedure

1. PSU off. Set the baseline jumpers and fit the cover.
2. Place and connect everything as in Figure 21, including the single plane → PSU ⏚ bond (check < 1 Ω to a scope BNC shell); the attenuators stay on.
3. Switch on: VOUT 12.2 V, LED on, input current ≈ 0.6 A.
4. Ambient: PSU off, remove the RG jumper (R8 holds the gate low, the converter stops), PSU on, record M1 and M2 as the noise floor.
5. PSU off, refit RG 1-2, PSU on, record the baseline CM and DM spectra and keep them as reference traces.
6. Change one knob at a time, always with the PSU off, and compare with the reference: Rg 4.7 → 22 → 100 Ω (CM and DM fall mainly in the 5–30 MHz band); SNUB on (20–30 MHz ringing peak lower); CASE = CHASSIS (CM up by 20–40 dB) versus PGND / 1 nF / floating; CM1 + CM2 (CM choke bypassed: CM up); DM (L2 bypassed: DM up, 150 kHz – 2 MHz); YCAP off (CM up); SGCH (earthed output: CM up) and then YPS 1-2 / 3-4 (CM down).

- **Cautions.** No passive probe on DRAIN, GATE or CS during the LISN measurement: its ground clip ties PGND to the scope earth and adds a CM path (use a differential probe for waveforms). The drain reaches ≈ 70 V and the Q1 tab is the drain: keep the cover on. The LED load dissipates ≈ 12 W.
- Near field: H-field probe over the hot loop and the case; RF current clamp on the J5 strap for the case and earthing knobs (expected $I_{CM}$ peaks: Section 6.3).

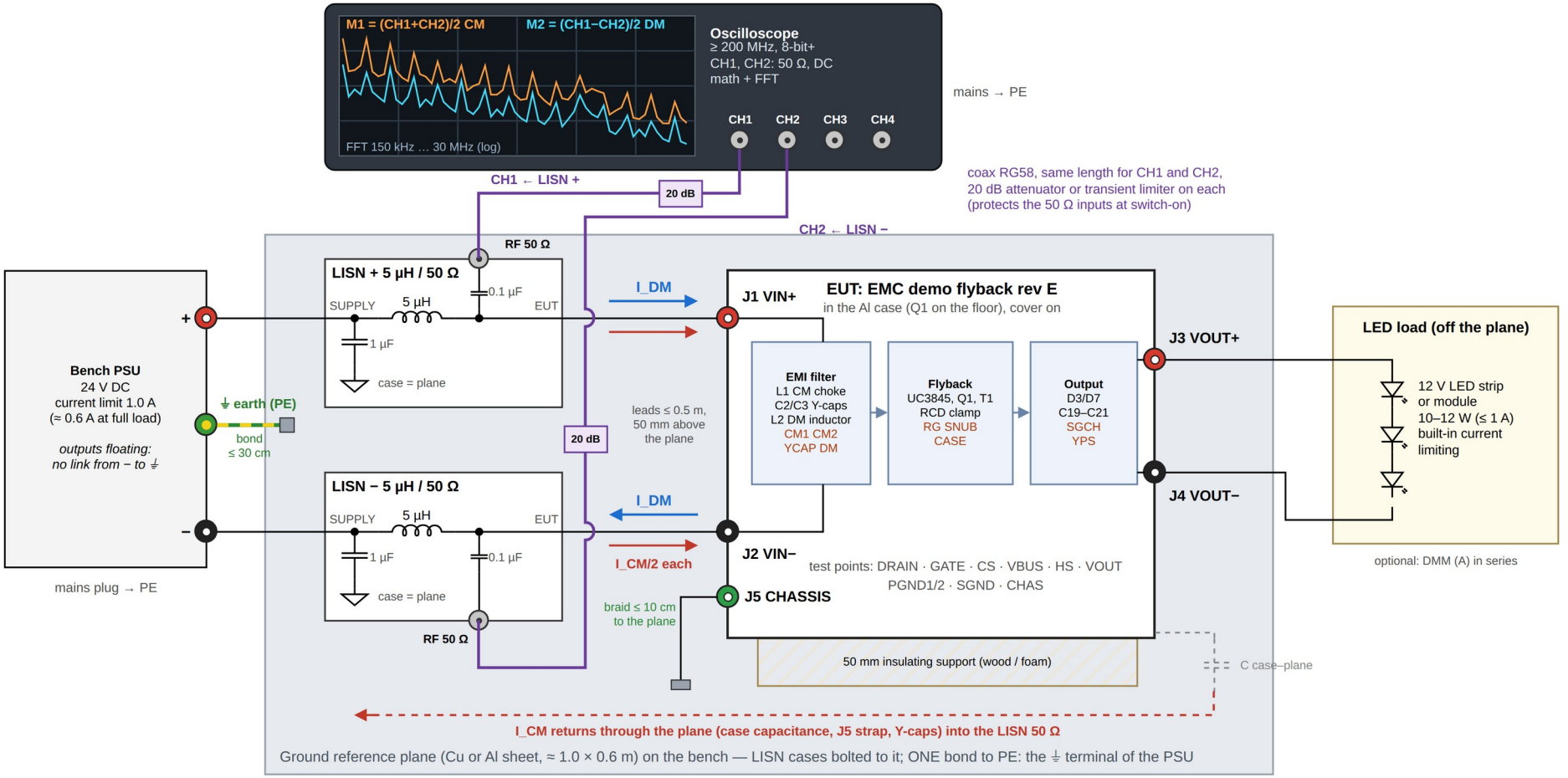



**Figure 21.** Conducted-emission test set-up: bench PSU → two 5 µH LISNs (cases bonded to the reference plane) → converter on a 50 mm support → LED load off the plane; the plane is bonded once to PE at the PSU ⏚ terminal. The LISN RF ports go through 20 dB attenuators to CH1/CH2 of the oscilloscope; scope math gives CM = (CH1+CH2)/2 and DM = (CH1−CH2)/2. Blue: differential-mode current; red: common-mode current returning through the plane into the LISN 50 Ω. Editable source: EMC_flyback_test_setup.drawio.

# 9 Risks and open points

- **Not yet prototyped.** Build one board first and measure all knobs before the lecture.
- **Models.** The controller, TL431 and opto models are behavioural (datasheet typical values); the COMP internal load $R_{div}$ is an assumption (30 kΩ; results barely change at 100 kΩ). Layout parasitics and common-mode paths are not in the simulation, so the predicted spectra are indicative only.
- **Output capacitor ripple current.** ≈ 1.9 $A_{rms}$ is shared between C18/C19 and C20/C21; check the ripple-current rating of RVT1E471M1010 and the temperature of C18/C19 at full load.
- **L2 stock.** C37429 (SWPA8040S100MT) is no longer listed in the LCSC shop; check JLC stock when ordering (any 10 µH, ≥ 3 A, 8 × 8 mm shielded inductor fits).
- **Assembly.** Check the JLC CPL rotations (U1, U2, U3, U4, diodes, LEDs, electrolytics) and the CM choke orientation (datasheet pin 1 at pad 1).
- **Q1 insulation and mounting.** The Sil-Pad is only 0.23 mm thick: deburr the floor and the tapped hole, check the insulation before soldering (Section 6.2), and do not over-tighten the screw. The leads are bent 3.8 mm from the body and leave about 1 mm reserve on the 12.7 mm minimum lead length; use a bending jig so that all five boards come out the same.
- **Cover temperature rise.** With the cover the board runs 13–17 K warmer (closed-cover upper bound, Section 6.2): clamp resistors up to ≈ 104 °C in the worst case. Measure through the vent slots on the first board; leave the cover off for long 100 Ω demonstrations in a warm room.
- **Case dimensions.** The drawing follows from the KiCad placement (make_case.py); if the PCB is changed, regenerate the drawing. Check the socket slots against the first case before ordering more.
- **Clamp-resistor and rectifier temperatures.** Rev E predicts ≈ 70 °C for R9/R10/R20 (≈ 88 °C worst case) and ≈ 51 °C for D3/D7, with the largest model uncertainty for the resistors (FR-4 spreading, up to +11 K). Measure them on the first board; C4 (electrolytic) sits next to the clamp resistors, so check its temperature for lifetime.
- **Paralleled Schottky diodes.** The forward voltage of a Schottky diode falls with temperature, so a hotter diode takes more current. D3 and D7 share copper and are thermally coupled (the model predicts < 1 K difference), but buy them from one reel and check at bring-up that both run at the same temperature. Two diodes also double the junction capacitance (≈ 2 × 300 pF) and the leakage, which slightly lowers the secondary ringing frequency.
- **LCSC number for R9/R10/R20.** The 4.7 kΩ 2512 1 W resistor (UNI-ROYAL 25121WJ0472T4E) is stocked by LCSC, but its C number could not be verified and is blank in JLC_BOM.csv. Fill it in from the JLC parts search when ordering.
- **No soft-start and no OVP.** Acceptable for a supervised lab demonstrator; add a series RC on COMP or a Zener clamp on VOUT if the board is used unsupervised.

# Appendix A  Design files

| File / folder | Content |
|---|---|
| emc_flyback (KiCad 7 project, opens in KiCad 7 and 8) | Schematic (with datasheet links), PCB rev E, 3D models in EMC_Demo.3dshapes |
| EMC_flyback_gerbers_JLC.zip, JLC_BOM.csv, JLC_CPL.csv | Fabrication and SMT assembly files |
| Full_BOM.csv, LCSC_hand_parts_order_5boards.csv | Complete BOM with datasheet and product-page links; hand-fitted parts order |
| EMC_flyback_topology.drawio (+ .png/.svg) | Editable topology drawing (Figure 1) |
| EMC_flyback_test_setup.drawio (+ .png/.svg) | Editable conducted-emission test set-up with procedure, baseline settings and cautions (Section 8.2) |
| simulation\EMC_flyback_power_stage.asc | LTspice open-loop power stage |
| simulation\EMC_flyback_closed_loop.asc, EMC_flyback_control.lib, *.asy* | LTspice closed loop with behavioural UC3845B/TL431/LTV-817 models and symbols |
| simulation\README_simulation.md | How to run, knobs, reference results |
| design_calc.py | Design calculations and loop-gain model used in Section 4 |
| thermal\extract2.py, thermal_model.py, run_revD.py | Board thermal model of Section 5.8 (copper extracted from the KiCad file; rev C version kept as thermal_model_revC.py) |
| case\EMC_flyback_case_revD_drawing.pdf, EMC_flyback_case_revD.step, EMC_flyback_cover_revD.step | Workshop drawing (page 1 case, page 2 cover) and 3D models of the case and the cover (generated by make_case.py from the PCB placement) |

# Appendix B  Bill of materials

| Ref | Value | MPN / description | LCSC | Assembly |
|---|---|---|---|---|
| J1 | VIN+ 24V | Cliff FCR7350R (red), 4 mm right-angle PCB socket | — | HAND |
| J2 | VIN- 0V | Cliff FCR7350B (black), 4 mm right-angle PCB socket | — | HAND |
| F1 | 2A/30V PTC | BSMD1812-200-30V | C960026 | SMT |
| D1 | SS510C | SS510C (reverse-polarity) | C19229 | SMT |
| L1 | 2.2mH WE-CMB | Wurth 744822222 (2.2 mH, 2 A) | — | HAND |
| CM1 | CM bypass + | PZ254V-11-02P | C492401 | HAND |
| CM2 | CM bypass - | PZ254V-11-02P | C492401 | HAND |
| C1 | 1u 50V | CL31B105KBHNNNE | C1848 | SMT |
| C2 | 10n 1kV | CC1206KKX7RCBB103 | C107192 | SMT |
| C3 | 10n 1kV | CC1206KKX7RCBB103 | C107192 | SMT |
| YCAP | Y-caps to chassis | PZ254V-11-02P | C492401 | HAND |
| L2 | 10uH 3.3A | SWPA8040S100MT | C37429 | SMT |
| DM | DM bypass | PZ254V-11-02P | C492401 | HAND |
| C4 | 100u 50V | MVK 50V100 M 10*10* | C165627 | SMT |
| C5 | 4.7u 50V | CL32B475KBUYNNE | C170099 | SMT |
| C6 | 4.7u 50V | CL32B475KBUYNNE | C170099 | SMT |
| R1 | 10k | 0603WAF1002T5E | C25804 | SMT |
| D5 | RED | NCD0805R1 | C84256 | SMT |

| Ref | Value | MPN / description | LCSC | Assembly |
|---|---|---|---|---|
| U2 | L78L12ACUTR | L78L12ACUTR | C69601 | SMT |
| C7 | 100n | CC0603KRX7R9BB104 | C14663 | SMT |
| C8 | 10u 25V | CL21A106KAYNNNE | C15850 | SMT |
| C9 | 100n | CC0603KRX7R9BB104 | C14663 | SMT |
| U1 | UC3845B | UC3845BD1013TR | C7118 | SMT |
| R2 | 5.6k | 0603WAF5601T5E | C23189 | SMT |
| C10 | 1n C0G | CL10C102JB8NNNC | C163508 | SMT |
| C11 | 100n | CC0603KRX7R9BB104 | C14663 | SMT |
| C12 | 1n | CL10B102KB8NNNC | C1588 | SMT |
| R3 | 1k | 0603WAF1001T5E | C21190 | SMT |
| C13 | 220p | CL10B221KB8NNNC | C1603 | SMT |
| R4 | 0.15R 2512 | HoLR2512-150mR-1% | C500726 | SMT |
| RG | Rg select | PZ254V-12-6P | C492420 | HAND |
| R5 | 4.7R | 0805W8F470KT5E | C17675 | SMT |
| R6 | 22R | 0805W8F220JT5E | C17561 | SMT |
| R7 | 100R | 0805W8F1000T5E | C17408 | SMT |
| R8 | 10k | 0603WAF1002T5E | C25804 | SMT |
| Q1 | IRF520NPBF | IRF520NPBF (Infineon) | C2598 | HAND |
| MH5 | M3 CASE | M3 x 8 mm metal hex stand-off (female-female) to the aluminium case | — | HAND |
| CASE | Case select | PZ254V-12-6P | C492420 | HAND |
| C14 | 1n 2kV | CC1206KKX7RDBB102 | C23631 | SMT |
| T1 | 750311591 | Wurth 750311591 (8 uH, 1.5:1:0.67, EFD25) | — | HAND |
| D2 | ES2D | ES2D | C364267 | SMT |
| C15 | 22n 100V | CC0805KRX7R0BB223 | C107137 | SMT |
| R9 | 4.7k 1W | 25121WJ0472T4E (UNI-ROYAL) | — | SMT |
| R10 | 4.7k 1W | 25121WJ0472T4E (UNI-ROYAL) | — | SMT |
| R20 | 4.7k 1W | 25121WJ0472T4E (UNI-ROYAL) | — | SMT |
| SNUB | RC snubber on | PZ254V-11-02P | C492401 | HAND |
| C16 | 100p 2kV C0G | TCC1206COG101J202DT | C2833637 | SMT |
| R11 | 47R | 0805W8F470JT5E | C17714 | SMT |
| D3 | SS510C | SS510C | C19229 | SMT |
| D7 | SS510C | SS510C | C19229 | SMT |
| R12 | 10R | 1206W4F100JT5E | C17903 | DNP |
| C17 | 100p 2kV | TCC1206COG101J202DT | C2833637 | DNP |
| C18 | 470u 25V | RVT1E471M1010 | C72518 | SMT |
| C19 | 470u 25V | RVT1E471M1010 | C72518 | SMT |
| C20 | 22u 25V | CL32A226KAJNNNE | C52306 | SMT |
| C21 | 22u 25V | CL32A226KAJNNNE | C52306 | SMT |
| R13 | 1k 1206 | 1206W4F1001T5E | C4410 | SMT |

| Ref | Value | MPN / description | LCSC | Assembly |
|---|---|---|---|---|
| R14 | 4.7k | 0603WAF4701T5E | C23162 | SMT |
| D6 | GREEN | KT-0805G | C2297 | SMT |
| J3 | VOUT +12V | Cliff FCR7350R (red), 4 mm right-angle PCB socket | — | HAND |
| J4 | VOUT 0V | Cliff FCR7350B (black), 4 mm right-angle PCB socket | — | HAND |
| R15 | 39k | 0603WAF3902T5E | C23153 | SMT |
| R16 | 10k | 0603WAF1002T5E | C25804 | SMT |
| U3 | TL431AIDBZR | TL431AIDBZR | C23892 | SMT |
| R17 | 1k | 0603WAF1001T5E | C21190 | SMT |
| R18 | 1k | 0603WAF1001T5E | C21190 | SMT |
| R19 | 10k | 0603WAF1002T5E | C25804 | SMT |
| C22 | 47n | CL10B473KB8NNNC | C1622 | SMT |
| U4 | LTV-817S | LTV-817S | C189217 | SMT |
| SGCH | SGND to chassis | PZ254V-11-02P | C492401 | HAND |
| YPS | Pri-sec Y-cap | PZ254V-12-4P | C492419 | HAND |
| C23 | 1n 2kV | CC1206KKX7RDBB102 | C23631 | SMT |
| C24 | 10n 1kV | CC1206KKX7RCBB103 | C107192 | SMT |
| J5 | CHASSIS | Cliff FCR7350G (green), 4 mm right-angle PCB socket | — | HAND |
| MH1 | M3 | | — | HAND |
| MH2 | M3 | | — | HAND |
| MH3 | M3 | | — | HAND |
| MH4 | M3 | | — | HAND |
| DRAIN | SW | Keystone 5002 (white) | C238123 | HAND |
| GATE | GATE | Keystone 5002 (white) | C238123 | HAND |
| CS | CSRAW | Keystone 5002 (white) | C238123 | HAND |
| PGND1 | PGND | Keystone 5001 (black) | C238122 | HAND |
| PGND2 | PGND | Keystone 5001 (black) | C238122 | HAND |
| VOUT | VOUT | Keystone 5002 (white) | C238123 | HAND |
| SGND | SGND | Keystone 5001 (black) | C238122 | HAND |
| HS | HS | Keystone 5002 (white) | C238123 | HAND |
| VBUS | VBUS | Keystone 5002 (white) | C238123 | HAND |
| CHAS | CHASSIS | Keystone 5001 (black) | C238122 | HAND |
| T1 | | Wurth 750311591 flyback transformer (EFD25, 8 uH, 1.5:1:0.67) | — | HAND |
| L1 | | Wurth 744822222 WE-CMB common-mode choke 2.2 mH / 2 A | — | HAND |
| Case | | Aluminium U-channel case = Q1 heatsink, 3 mm 5754/5052, 112 x 94 x 33 mm (drawing EMC_flyback_c | — | HAND |
| Q1 pad | | Bergquist Sil-Pad 400 TO-220 pre-cut, 0.009 in (SP400-0.009-00-54) - insulator between Q1 tab a | — | HAND |
| Q1 washer | | TO-220 nylon shoulder washer for M3 (e.g. Keystone 3049) | — | HAND |
| Q1 screw | | M3 x 6 pan-head steel screw (into the tapped case floor) + M3 flat washer | — | HAND |
| J1,J3 | | Cliff FCR7350R 4 mm right-angle PCB socket, red | — | HAND |

| Ref | Value | MPN / description | LCSC | Assembly |
|---|---|---|---|---|
| J2,J4 | | Cliff FCR7350B 4 mm right-angle PCB socket, black | — | HAND |
| J5 | | Cliff FCR7350G 4 mm right-angle PCB socket, green (chassis/earth) | — | HAND |
| MH1-MH4 | | M3 x 8 NYLON hex stand-off F-F + M3 x 6 nylon pan screw (top) + M3 x 6 countersunk screw (botto | — | HAND |
| MH5 | | M3 x 8 METAL hex stand-off F-F + M3 x 6 steel pan screw with washer (top) + M3 x 6 countersunk | — | HAND |
| Cover | | Transparent cover, 3 mm clear cast PMMA (acrylic), 112 x 94 mm, laser cut with 4 holes and 5 ve | — | HAND |
| Cover pillars | | M3 x 30 aluminium hex stand-off F-F + M3 x 6 countersunk screw (floor) + M3 x 8 nylon pan screw | — | HAND |
| Feet | | Self-adhesive rubber feet, >= 3 mm high (case must not touch the ground plate when CASE is open | — | HAND |
| Chassis | | Aluminium ground plate ~300x200x2 mm (reference ground / 'metal housing') | — | HAND |
| Jumpers | | 2.54 mm jumper shunts (TE 2-382811-1, LCSC C86478) | C86478 | HAND |